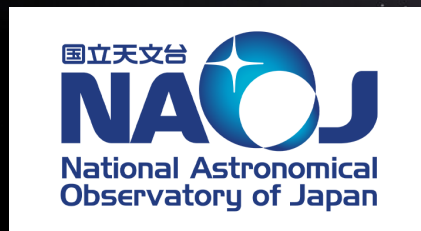


Polarimetric observations of the NEA “2024 MK”
during its close approach within Lunar Distance

Photometric observations of the
Hayabusa2# Flyby target NEA “Torifune”



EOU Seminar
22 Dec. 2025 14:45-16:15
M2 Kazuya DOI

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1-1. Classification of Asteroids

【Classification by Orbit】

Yellow line: Jupiter

Red line: Mars

White line: Earth

Yellow points

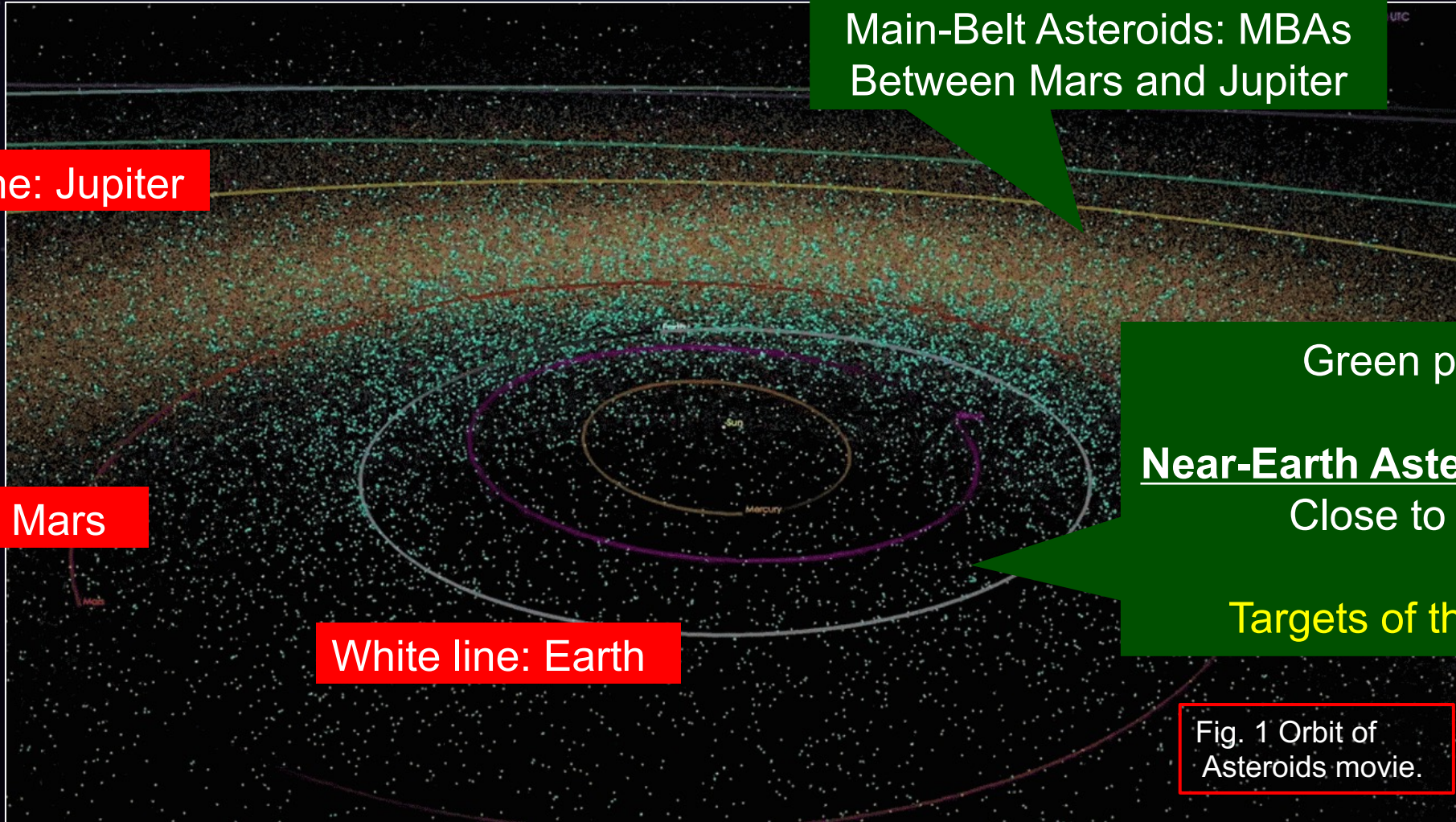
Main-Belt Asteroids: MBAs
Between Mars and Jupiter

Green points

Near-Earth Asteroids: NEAs
Close to Earth

Targets of this study!

Fig. 1 Orbit of
Asteroids movie.



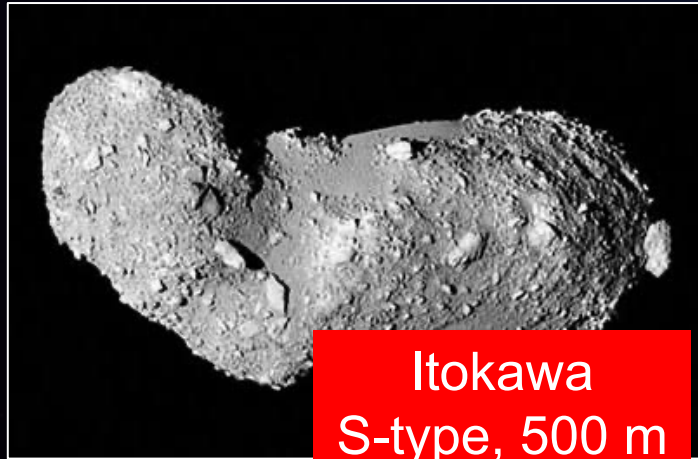
1-1. Classification of Asteroids

【Potentially-Hazardous Asteroids: PHAs】

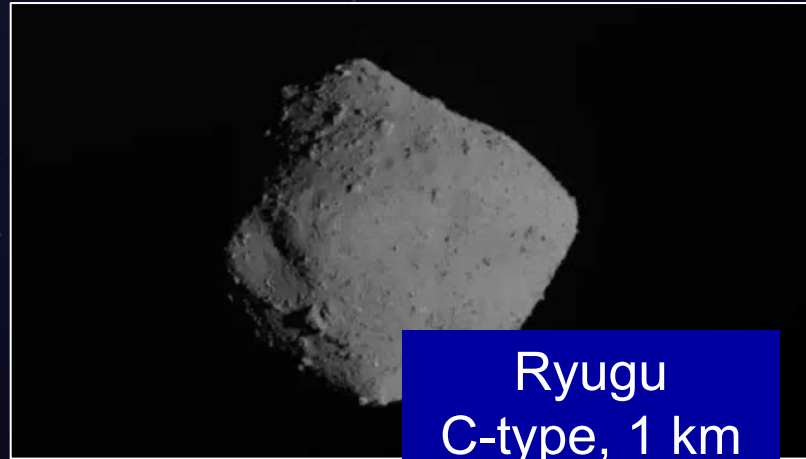
Near & Large



- Specifically, minimum orbit intersection distance is within 0.05 au, and diameter > 140 m NEAs.
- PHAs have a high possibility of collision, and cause significant potential hazards.
e.g., Itokawa, Ryugu, Apophis



Itokawa
S-type, 500 m
Albedo ~0.25



Ryugu
C-type, 1 km
Albedo ~0.05



Apophis
S (Sq)-type,
340 m

Fig. 2, 3, 4, 5 Image of asteroid impact, S-type PHA Itokawa, C-type PHA Ryugu, Orbit of PHA Apophis.

1-1. Classification of Asteroids

【Classification by Composition: Taxonomy】

5 major types

1. **S-type**

Stony, silicate and olivine-rich.

2. **C-type**

Carbonaceous.

3. **D-type**

Tagish Lake meteorites.

4. **V-type**

Basaltic, fragments of Vesta.

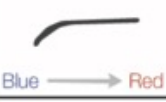
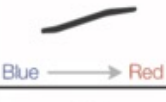
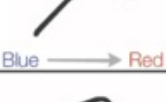
5. **X-type**

Various metallic meteorites.



Fig. 6, 7 S-type Asteroid Itokawa, C-type Asteroid Ryugu.

Tab. 1 Characteristics of the five major types.

Major Taxonomic Types	Reflectance Spectrum (0.4-0.9 μm)	Spectral Features	Visible Albedo	Suspected Composition
D (D,T)		Relatively featureless spectrum Steep red slope	0.02-0.06	Primitive carbonaceous Organic-rich compounds Hydrated minerals
C (C,B,F,G)		Slight bluish to slight reddish slope Shallow to deep absorption blueward of 0.5 μm Hydrated asteroids with absorption at 0.7 and 3.0 μm	<u>0.03-0.10</u>	Hydrated minerals Silicates Organics
X (E,M,P)		Slightly reddish spectrum E: absorption features at 0.5 and 0.6 μm	E: 0.18-0.4 M: 0.10-0.18 P: 0.03-0.10	
S (S,Q,A,K,L)		Moderately steep red slope ($\lambda < 0.7 \mu\text{m}$) Shallow to deep absorption at 1.0 and 2.0 μm	<u>0.10-0.22</u>	Stony composition Magnesium Iron silicates
V		Moderate to steep red slope Very deep absorption at 0.9 μm	0.05-0.60	Volcanic basalts Plutonic rocks

C-type
Low albedo

S-type
Moderate albedo

1-2. Origin of NEAs

【Asteroids as survivors of planetesimals】

- Asteroids retain primordial information from before the formation of planets.

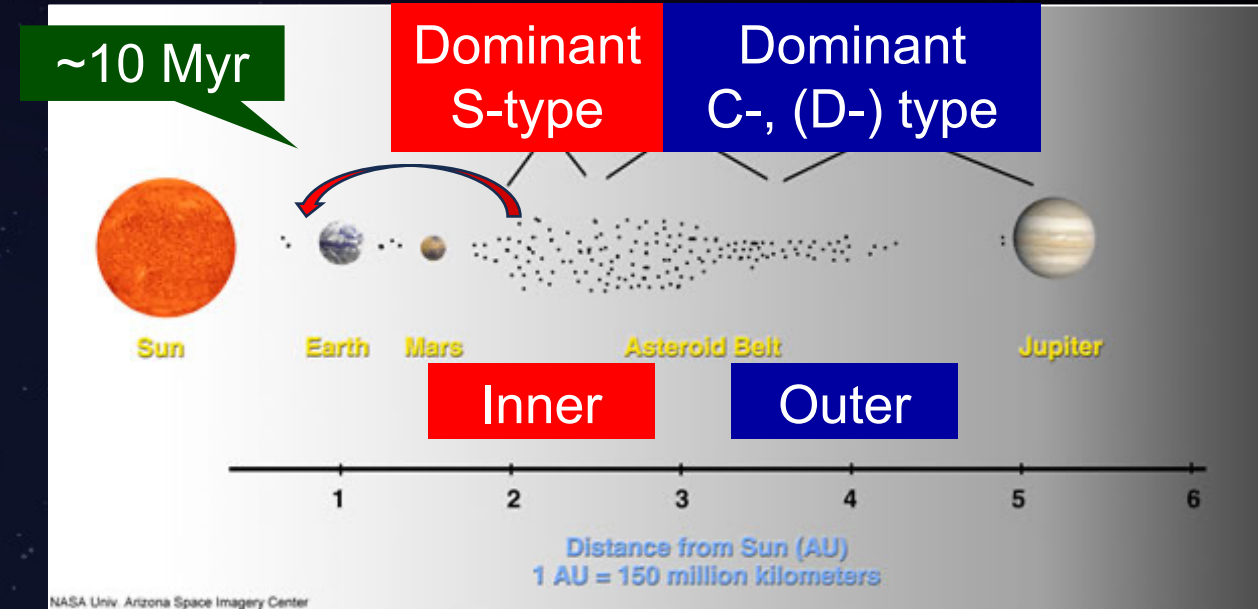


Fig. 8 Simple model of distribution of asteroids in the Solar System.

【Most NEAs originate from the Inner Main-belt】

- Scattered by Jupiter's gravity, many S-types (stony) come from the inner Main-belt.
- In contrast, C-types (carbonaceous) are primarily located in the outer Main-belt.

1-3. Number

【1.4 million discovered (as of 2025)】

- The number of discoveries is increasing steeply.
- Due to improvements of observational instruments and skills.

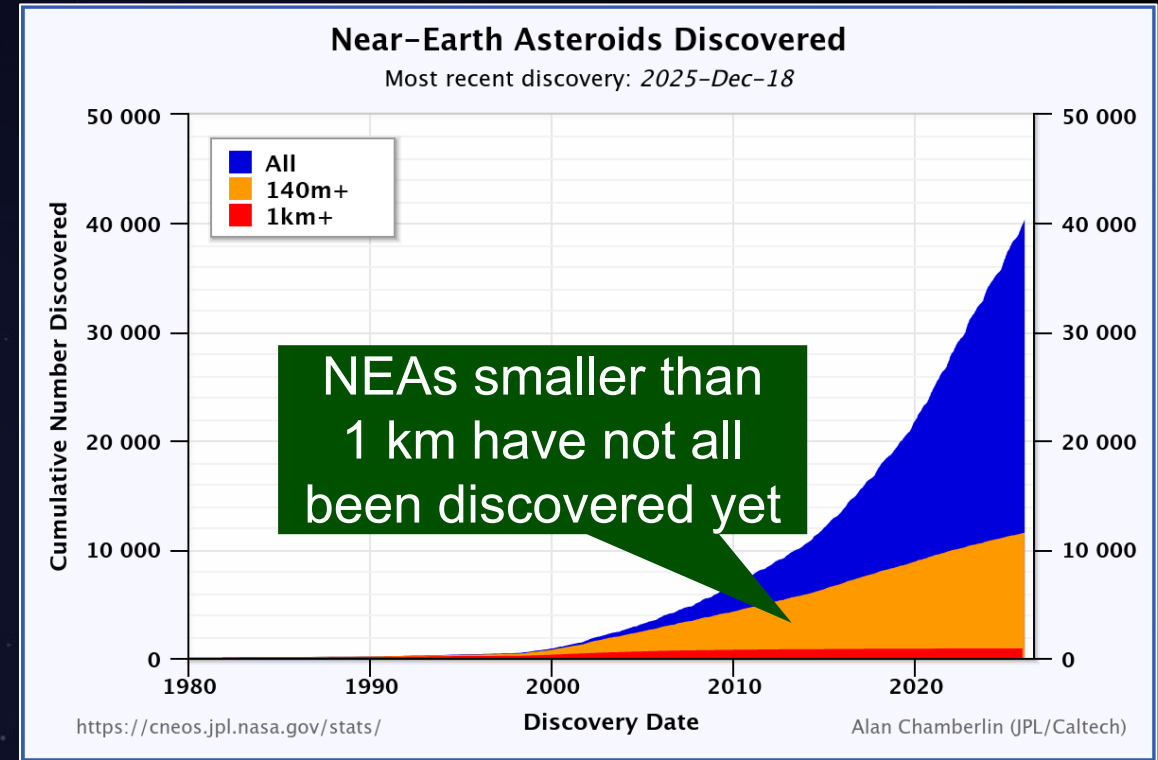


Fig. 9 Number of NEAs discovered as of 2025.

【Number of NEAs (as of 2025)】

- ~40,000 have been discovered, but the physical properties of many newly discovered NEAs are unknown.
- Most NEAs larger than 1 km have been discovered because they are bright, but many NEAs smaller than 1 km remain undiscovered because they are dark.

We can not respond to the unknown!

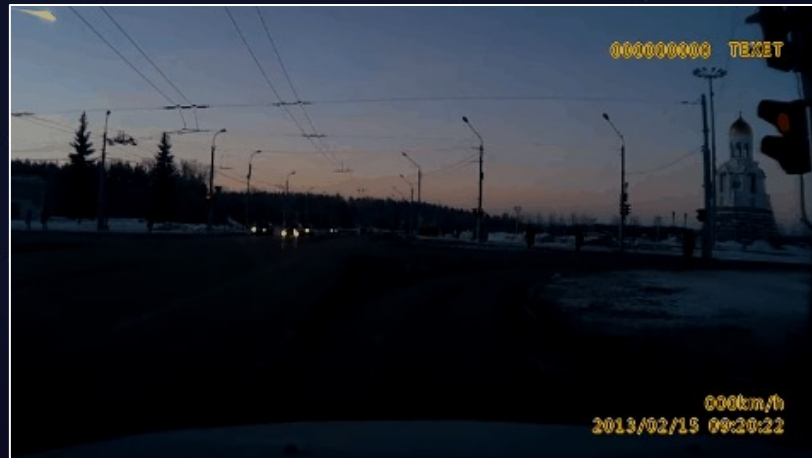
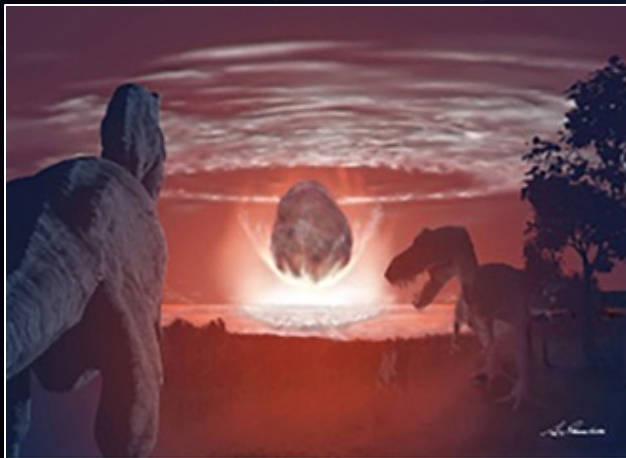
1-4. Planetary Defense

Avoiding collisions

Predicting hazard

【Protecting the Earth from Asteroids (and Comets) collisions】

- 66 million years ago, Dinosaurs extinction was caused by collision of 10-20 km Asteroid (or Comet).
- In 2013, ~20 m Asteroid exploded over Chelyabinsk, injuring 1,500 people.
- In 1994, Comet Shoemaker-Levy 9 collided with Jupiter. etc.



- So now, multiple missions are being conducted to achieve and contribute Planetary Defense by NASA, ESA and JAXA etc.

Fig. 10, 11, 12 Chicxulub meteor, Chelyabinsk meteor explosion movie, The collision of Comet Shoemaker-Levy 9 with Jupiter.

1-4. Planetary Defense

【Planetary Defense response process】

Step1: Discovery

Orbit determination through follow-up observations

Step2: Investigate of **physical properties**

(size, composition, porosity etc.)

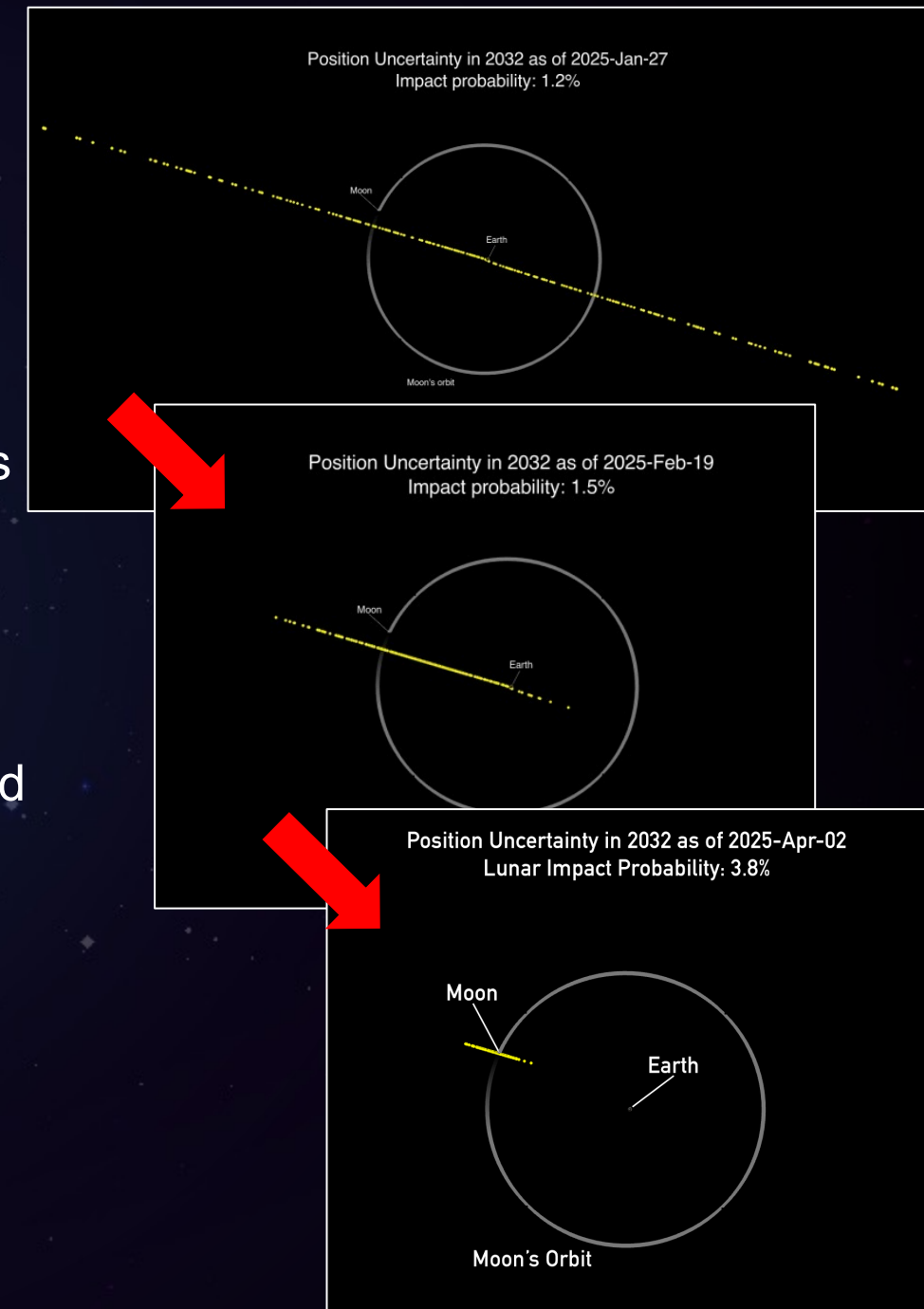
To evaluate the possibility of changing the orbit and predicting hazards

This study contributes to this step!

Step3: Avoidance or Damage minimization

Spacecraft impact, Alert to the public etc.

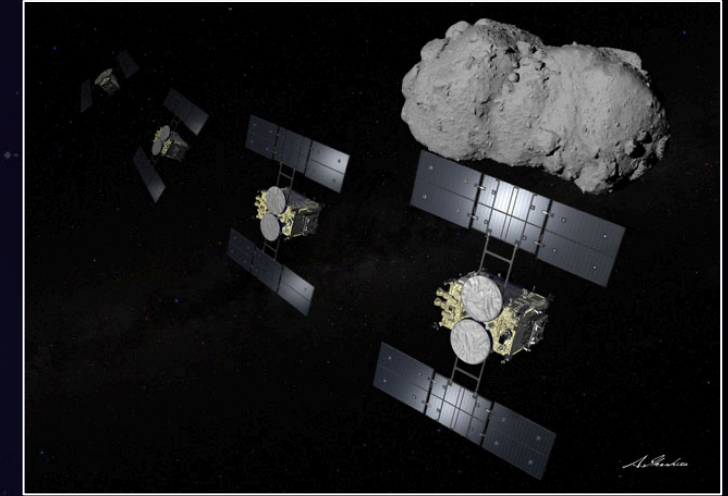
Fig. 13 Change in position uncertainty of NEA 2024 YR4.



1-5. JAXA Hayabusa2# Mission

【Hayabusa2# Mission objectives】

1. Advancement of long-term solar system navigation technology.
2. Realization of exploration of small and fast rotating Asteroids.
3. Acquisition of science and technology contributing to Planetary Defense.



【Targets Asteroids】

1. Torifune
Flyby in 5 July 2026. PHA, Itokawa like?
2. 1998 KY26
Rendezvous in July 2031.
Size: <17 m, Rot. Period: 10 min (so fast). Monolith?

This study
observed Torifune

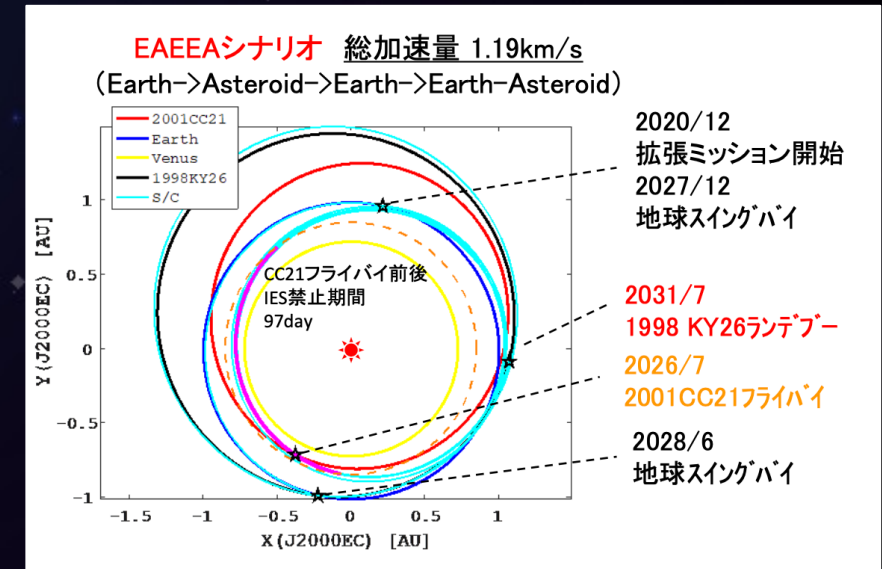


Fig. 14, 15 Hayabusa2 Flyby image, JAXA EAAEA scenario.

1-5. JAXA Hayabusa2# Mission

【Torifune Flyby】

- Flyby speed is ~5 km/s (so fast!)
- Planned closest distance ~1 km (“Kamihitoe method”)

【Hayabusa2 onboard camera】

- Designed for rendezvous (has not zoom function), so it is not capable of telephoto imaging.

- At the Flyby, the exposure time and gain must be pre-determined.

- Surface color information from ground-based observations in advance is essential.

This study!

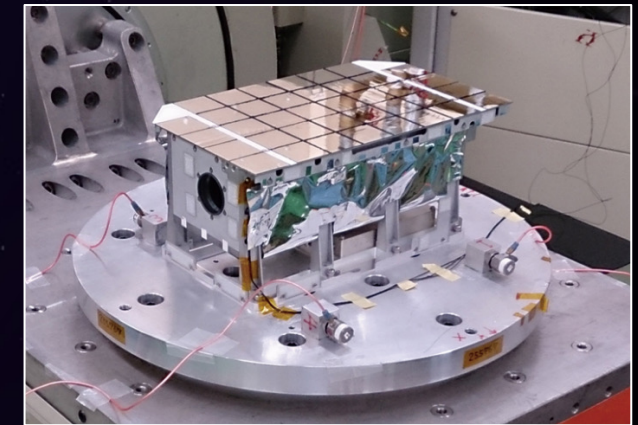
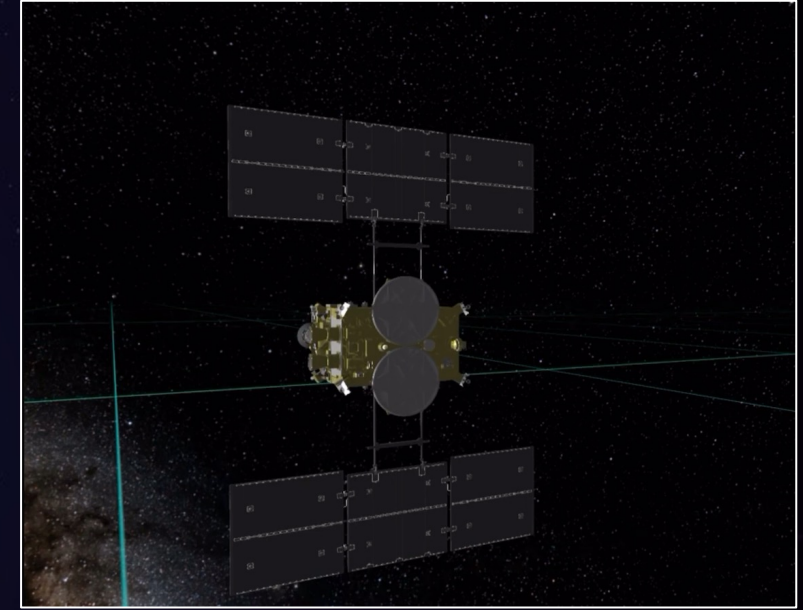


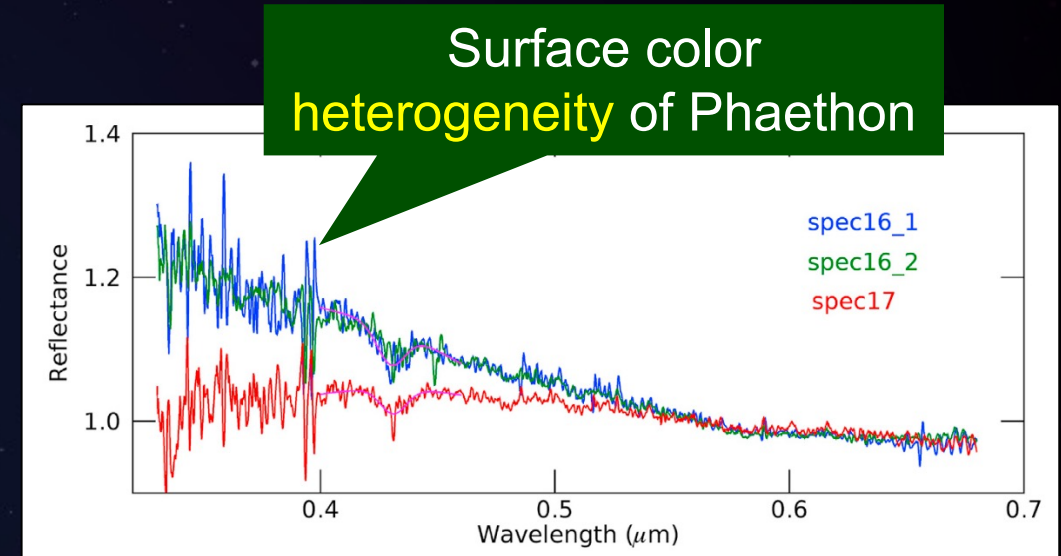
Fig. 16, 17 Simulation of the Torifune Flyby at ~1 km, Hayabusa2's onboard NIRS3 spectroscopic camera.

1-5. JAXA Hayabusa2# Mission

【Surface color homogeneity or heterogeneity】

- Many asteroids have been confirmed to exhibit surface color homogeneity.

Some asteroids have been reported to exhibit surface color heterogeneity (front-back side).
e.g., NEA Phaethon



【Flyby can observe only one (front) side】

- Ground-based observations can provide color information on the most scientifically valuable side.

Spatially resolvable time is only 2-3 s!

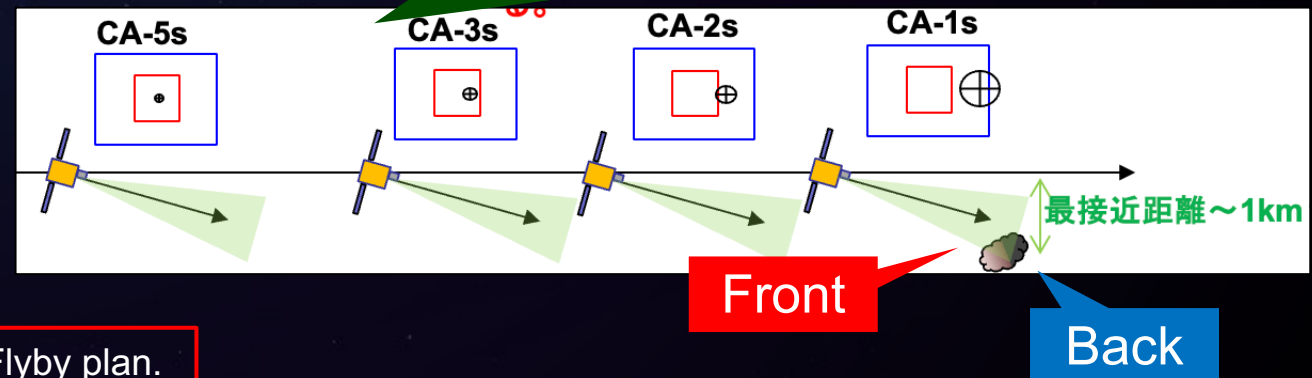


Fig. 18, 19 Reflectance spectrum of Phaethon, Hayabusa2 Flyby plan.

2. Purpose and Goals

【Summary of Introduction】

- Clarifying NEAs is so important; Planetary Defense, Surface information etc.
- Ground-based observations before the “in-situ” observations are essential.

【Goals】

- Obtaining information (behavior) to understand NEAs.
- Contributing to Planetary Defense, especially Hayabusa2# mission.

【Purpose】

- Deriving the physical properties of unique NEAs through ground-based observations. surface composition, albedo, grain size and porosity etc. (Representative NEA)

3-1. Observation methods

(A) Polarimetric observations

- Grain size
- Porosity
- Surface composition (rough)
- Albedo

Pirka

(B) Photometric observations

- (Local) Surface composition
- Size
- Rotation period
- Shape

Seimei

Spectroscopic observations

- Surface composition

Occultation observations

- Orbit
- Size
- Shape

Radar observations

- Size
- Shape

Other observations

(Thermal infrared, Survey, In situ, Astrometry etc.)

- Temperature
- Orbit
- Size
- Shape
- Albedo
- Rotation Period
- Albedo etc.

3-2. Identification of target Asteroids

【Identification method】

- 'Blink' 2 fits data (images taken 5-10 min apart), while tracking in Asteroid (non-sidereal) mode, so that the moving objects are stars (sidereal), and the stationary object is the target Asteroids.

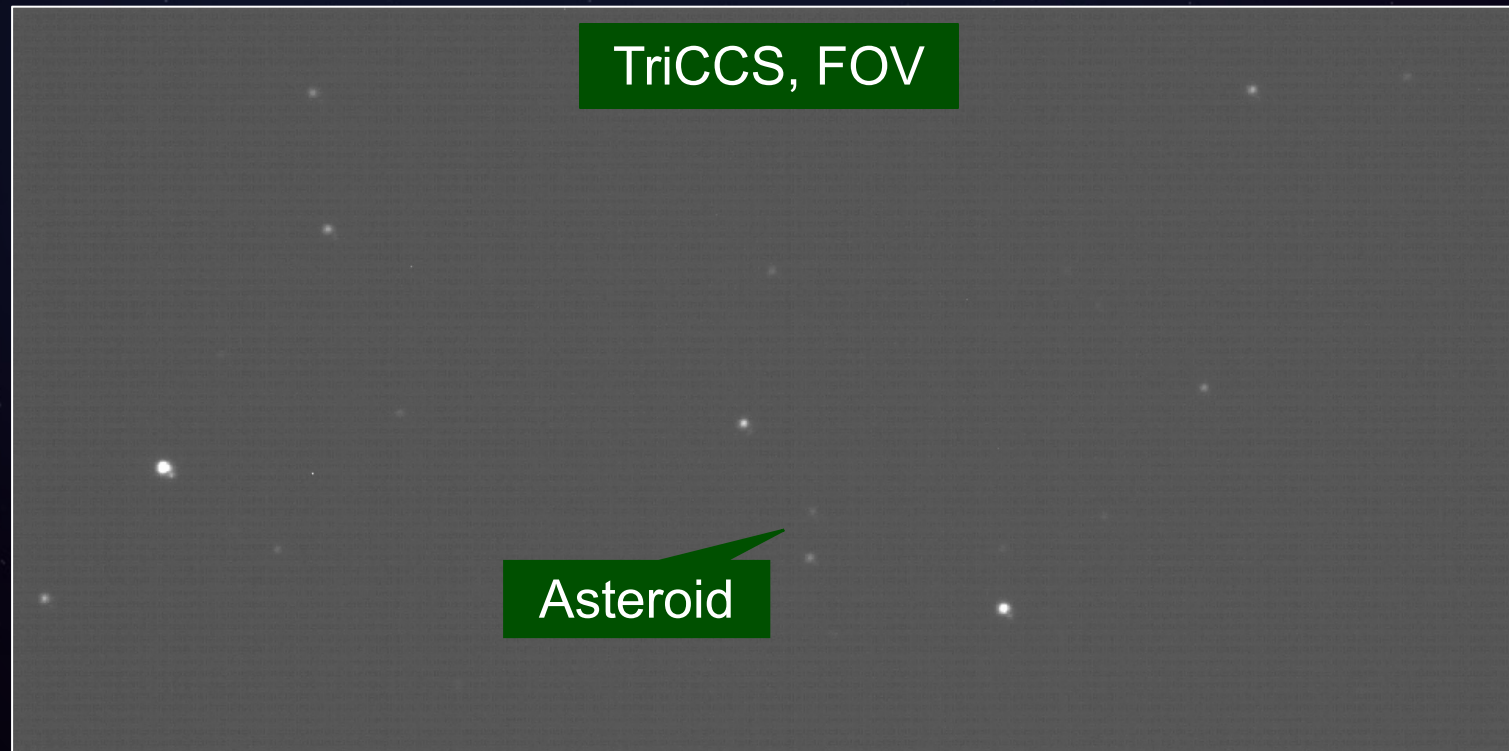


Fig. 20 2 fits data of target Asteroids blink movie.

4-1. Polarimetric observations

【Polarization】

- Light that vibrates in various directions, with only certain directions of vibration being selected.

Asteroids reflect sunlight (unpolarized light) and polarization occurs depending on the surface scattering properties.



Unpolarized & Polarized light

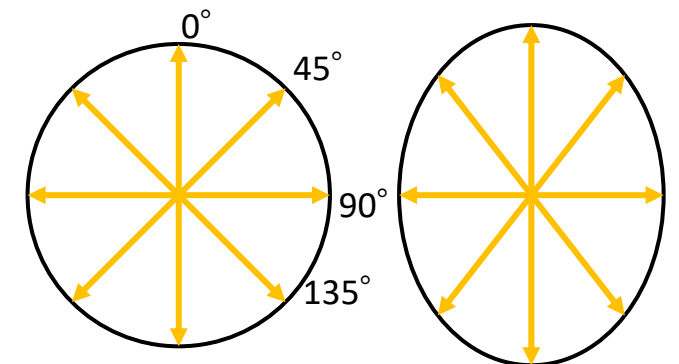


Fig. 21, 22, 23 Polarization filter movie, Polarization of Asteroid, Unpolarized and polarized light.

4-1. Polarimetric observations

【Polarization degree】

- Linear polarization degree: $P_r = \frac{I_{\perp} - I_{\parallel}}{I_{\perp} + I_{\parallel}} \%$
- Polarizer: Wollaston prism
separate into Ordinary-ray (O-ray: $I_{\perp}$) and Extraordinary-ray (E-ray: $I_{\parallel}$)

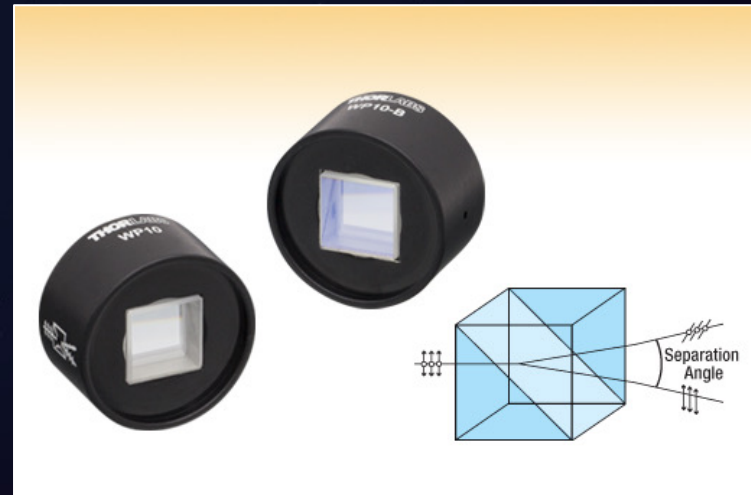
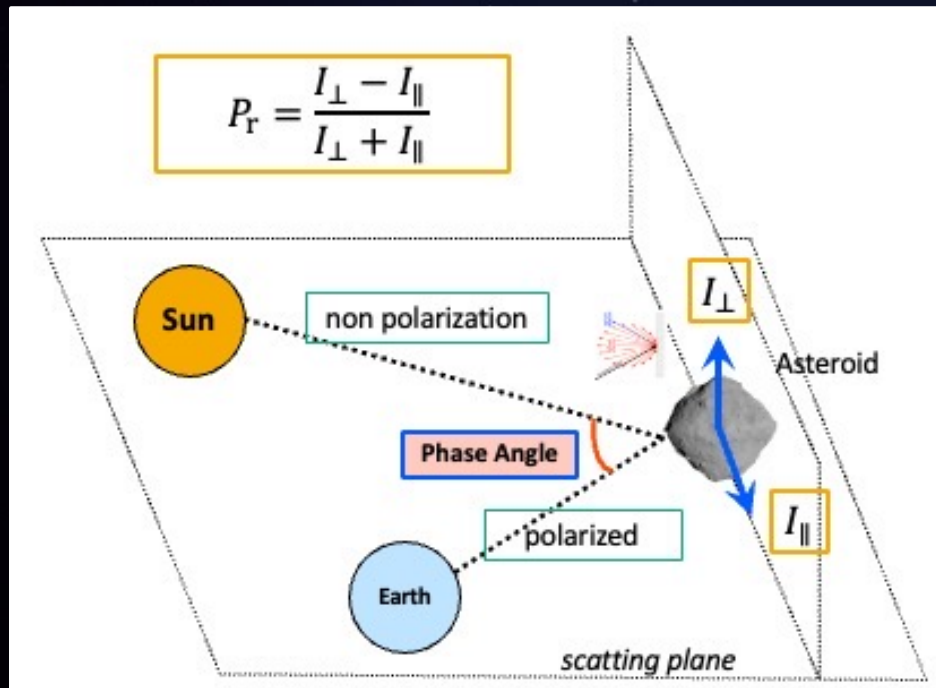
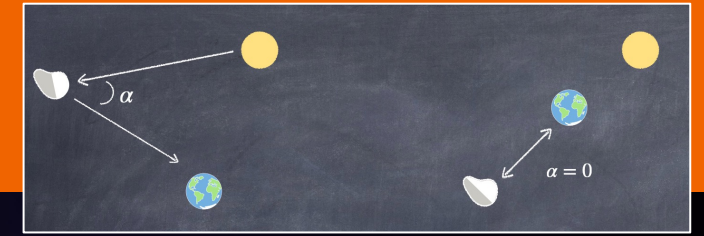


Fig. 24, 25, 26 Basic information of asteroid polarization, Wollaston Prism, MSI FOV.

4-1. Polarimetric observations



【Polarization degree】

- Linear polarization degree: P_r depends on the phase angle: α

【Surface scattering properties】

- Surface composition & Albedo**
Polarization curve shape (visually)
estimate Taxonomy type (C or S or E(X))

Tab. 2 Polarization parameter characteristics of C-type and S-type.

Polarization characteristics	α_{inv}	h	P_{max}	P_{min}
C-type	Small	Steep	High	Deep
S-type	Large	Mild	Low	Shallow

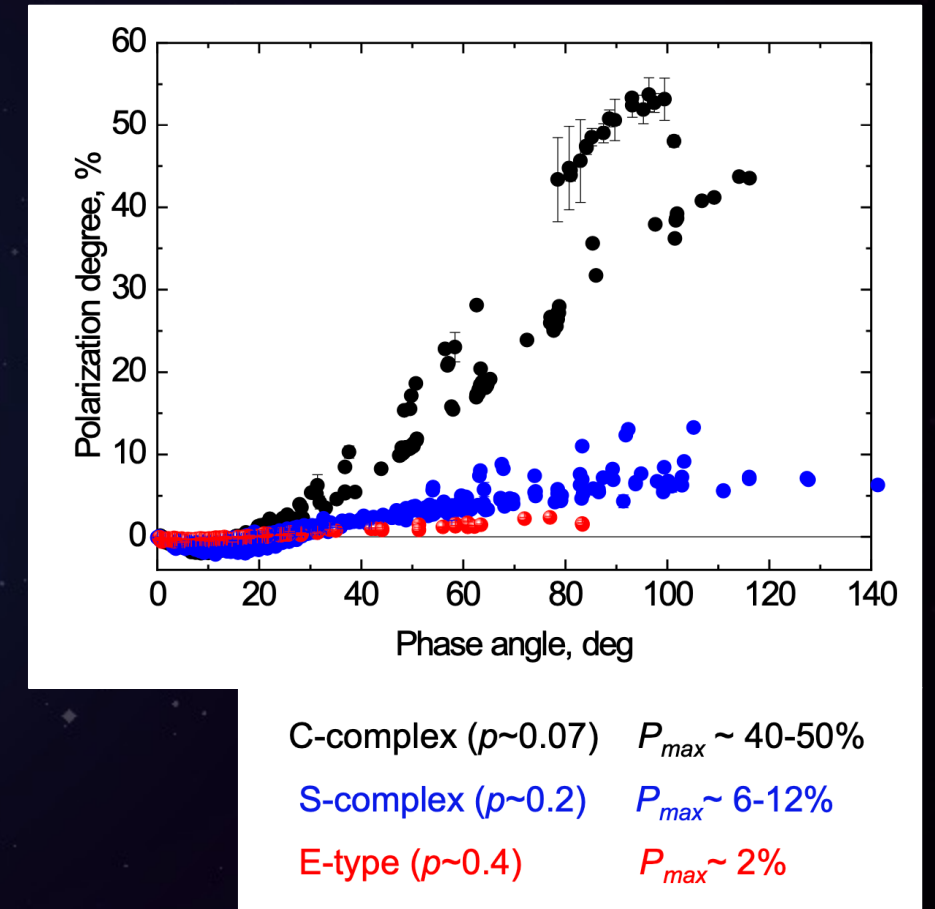


Fig. 27, 28 Phase angle, Polarization of typical C-, S-, E-Type Asteroids include NEAs [Belskaya, 2024].

4-1. Polarimetric observations

【Surface scattering property】

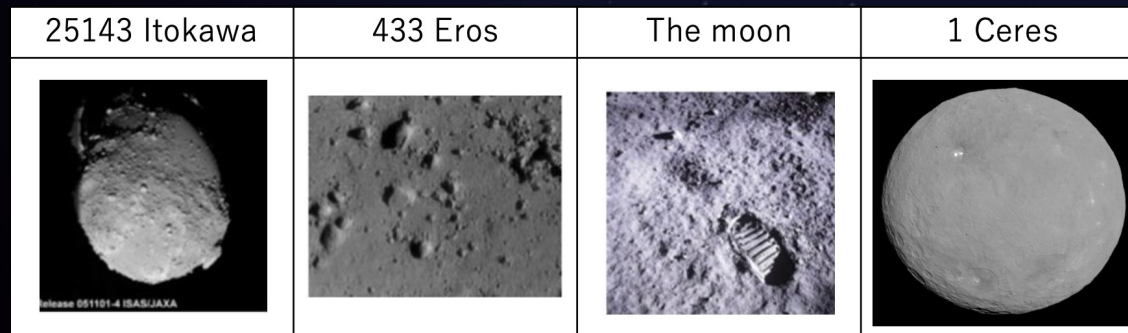
- Regolith grain size

Laboratory experiments, the Linear polarization degree tends to increase when the particle size is either much smaller or larger than the observed wavelength.

Grain size parameter: $X = \frac{\pi d}{\lambda}$

d : Grain diameter [μm]

λ : Wavelength of observational light [μm]



Rough

Grain size

Fine

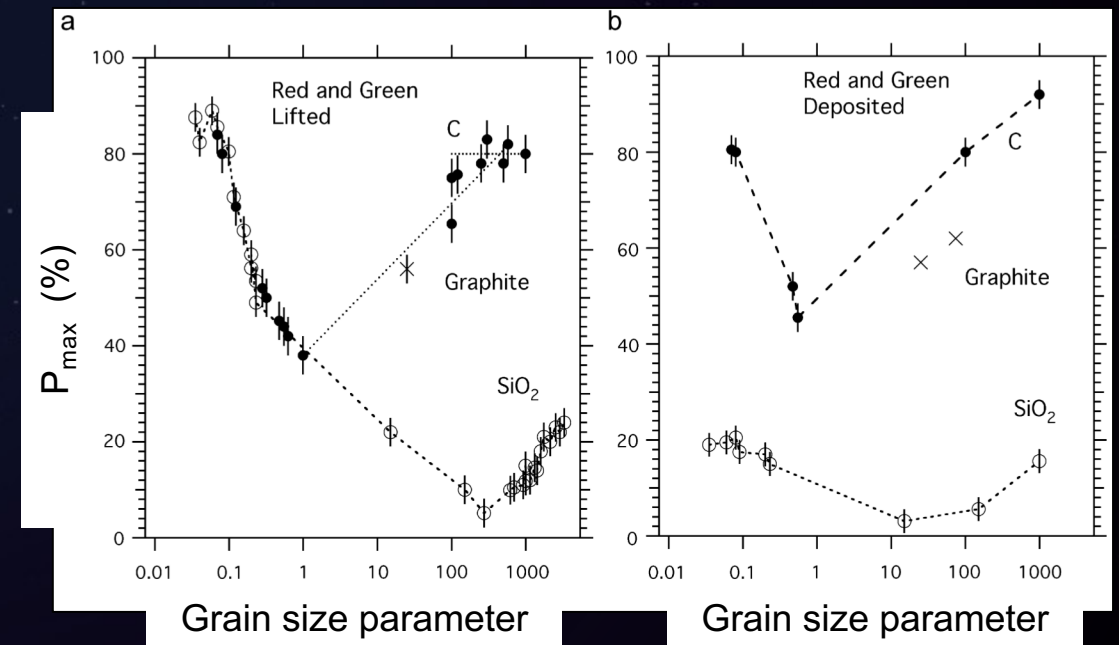


Fig. 29, 30 Regolith grain size of some asteroids and moon, P_{max} vs Grain size parameter [Hadamcik et al., 2009].

4-1. Polarimetric observations

【Surface scattering property】

- **Porosity**

Regardless of grain size, the Liner polarization degree tends to increase strongly due to high porosity arising from irregular grain shapes and heterogeneous surface structures.

NEA Phaethon has shown higher than expected polarization degrees and such behavior has been attributed to highly porous surface.

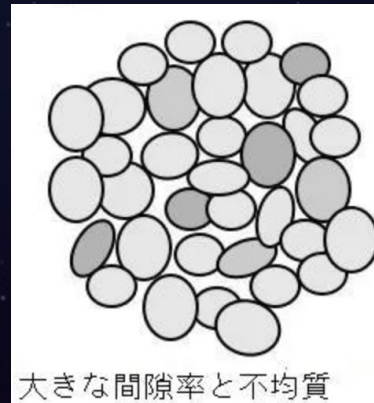
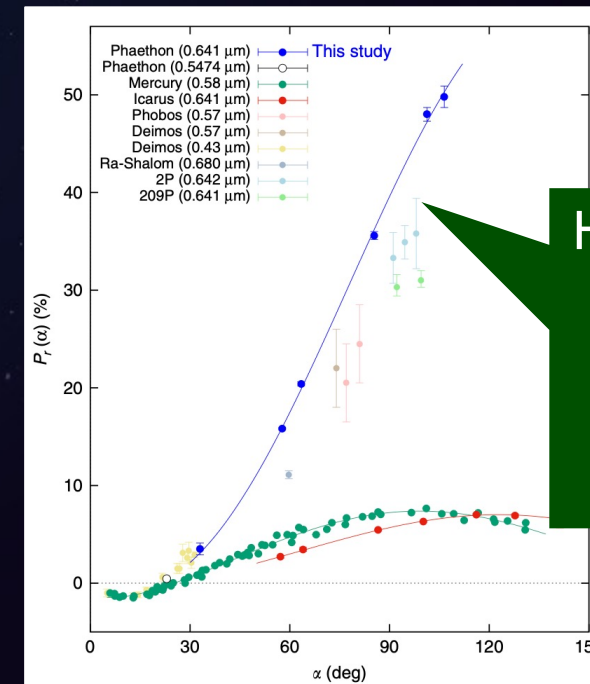


Fig. 31, 32 Image of high porous, Liner Polarization degree of NEA Phaethon [Ito et al., 2018].



Higher polarization
than expected

Because of the
high porosity

4-1. Polarimetric observations

【Surface scattering property】

- **Surface composition & Albedo**
- **Regolith grain size**
- **Porosity**



- The surface scattering properties are not simple, as they involve several contributing factors.
- It is certain that the polarization degree of Asteroids depends on surface scattering properties, but the detail remain unclear.

4-2. 1.6 m Pirka Telescope & MSI

【Pirka Telescope】

- Location: Nayoro Observatory
- Own: Hokkaido univ.

【MSI (Multi-Spectral Imager; Watanabe et al., 2012)】

- Wavelength: $0.36\text{--}1.05\ \mu\text{m}$
- Field of view (FOV): $3.3' \times 3.3'$
- Array format: 512×512 pixels
- Filter: Johnson-Cousins U, B, V, R_c, I_c
- Polarizer: Wollaston Prism

Can conduct polarimetry

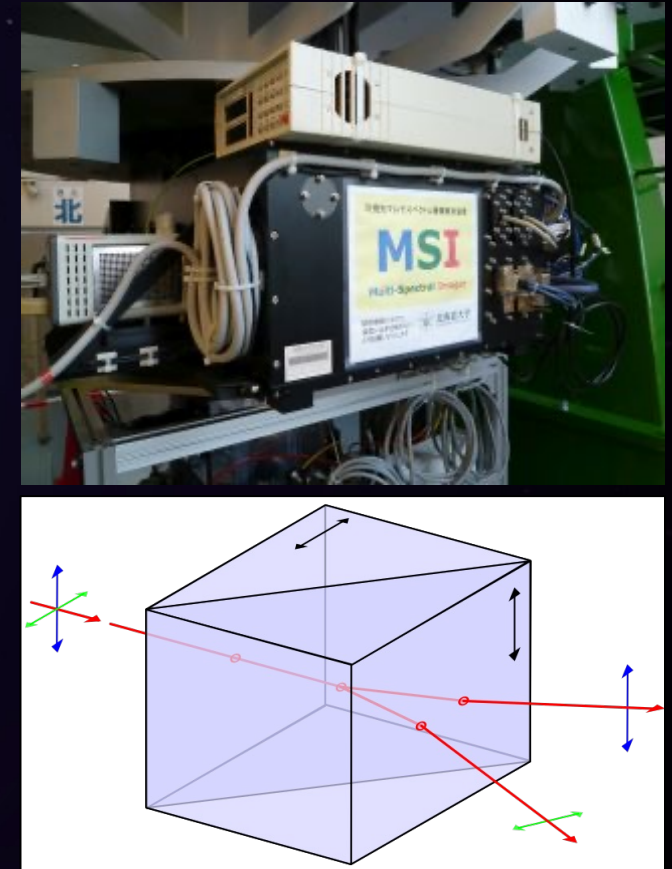
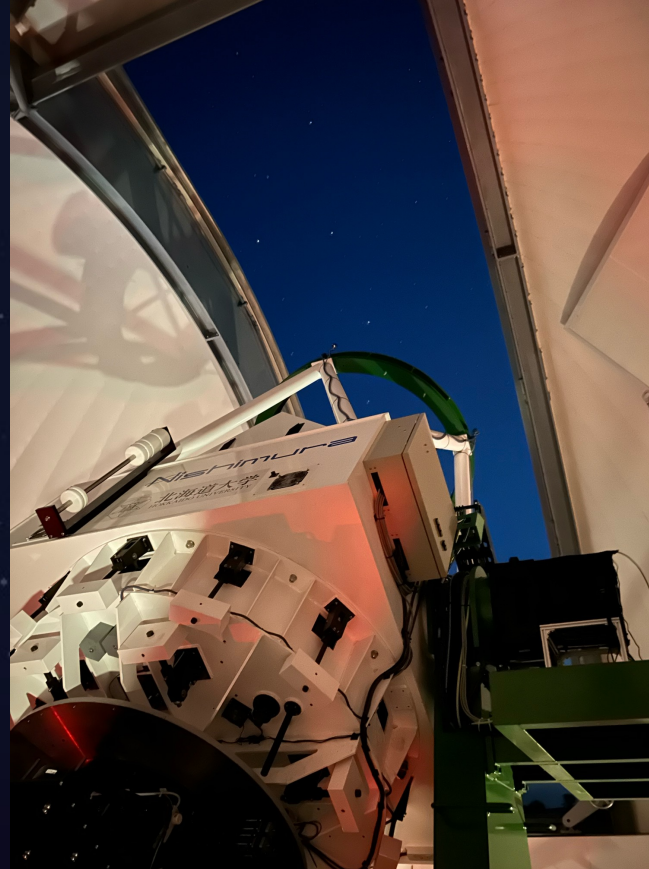


Fig. 33, 34, 35 Pirka Telescope, MSI [Watanabe et al., 2012], Wollaston prism.

4-3. 2024 MK Information

【Information】

- Discovered at 16 June 2024, closest approach within the ~76 % Lunar Distance (LD) at 29 June 2024 (only once in several decades event!).
- Minimum V-mag. over the next ~200 years: 17.8 mag., only opportunity for ground -based observations (mid-size telescope).



Fig. 36 Orbit around closest approach day of 2024 MK.

Tab. 3 2024 MK; 6th closest approach among known PHAs.

Year	Name	Geocentric distance		Estimate size [m]
		[km]	[LD]	
1957	2019 CD2	75300	0.20	260-590
2001	2017 VW13	120200	0.31	200-440
1914	1998 KJ9	233200	0.61	280-900
1971	2002 JE9	263600	0.69	140-310
1923	2021 MK1	273500	0.71	140-320
2024	2024 MK	295400	0.76	100-230
2018	2018 AH	297000	0.77	80-170

4-3. 2024 MK Information

【Other studies】

- NASA Goldstone Radar observations derived shape (meter-scale topography including hills, ridges, concavities and boulders) and size ~ 140 m (Naidu et al., 2024).

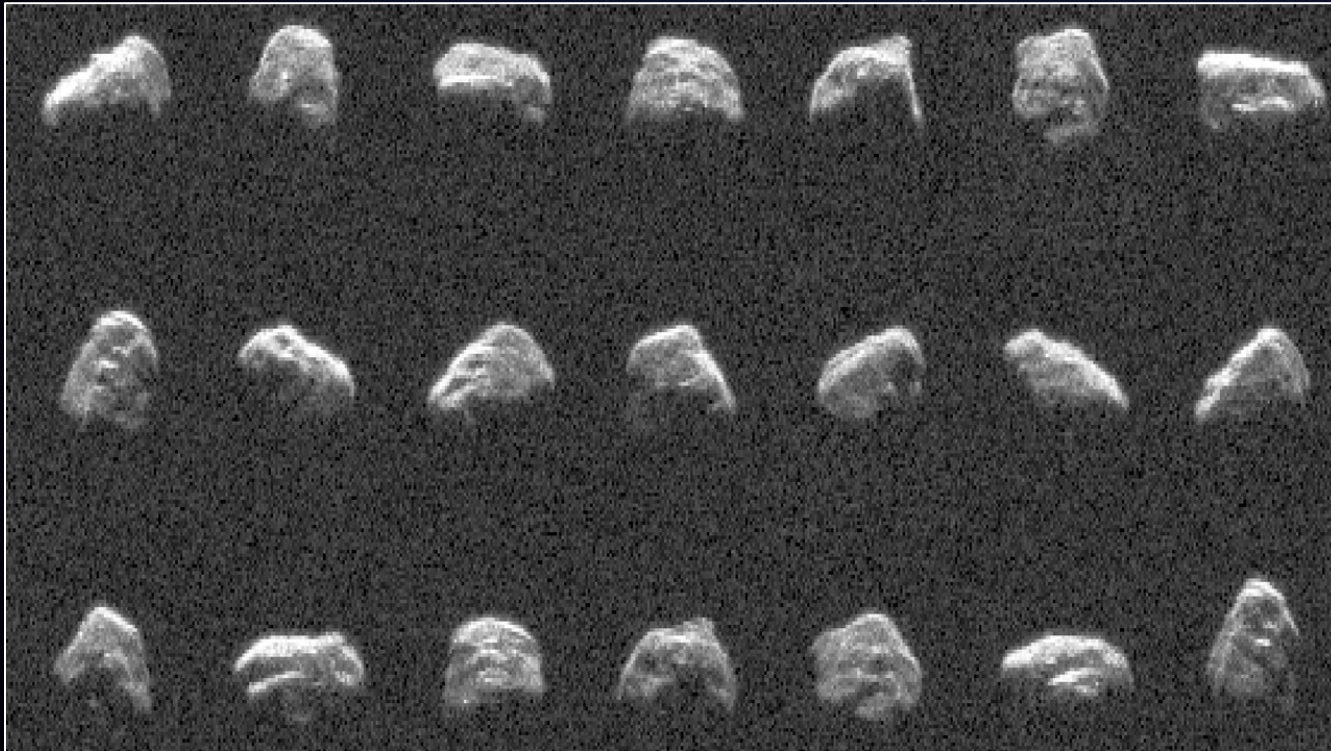


Fig. 37, 38 NASA Goldstone radar images and movie of 2024 MK [Naidu et al., 2024].

4-3. 2024 MK Information

【Other studies】

- NASA IRTF/SpeX Spectroscopic observations derived reflectance spectrum most likely **S-type** (McGraw et al., 2024).
- Light-curve observations derived rotation period 0.66 and/or 0.73 hours (Sioulas, 2024).

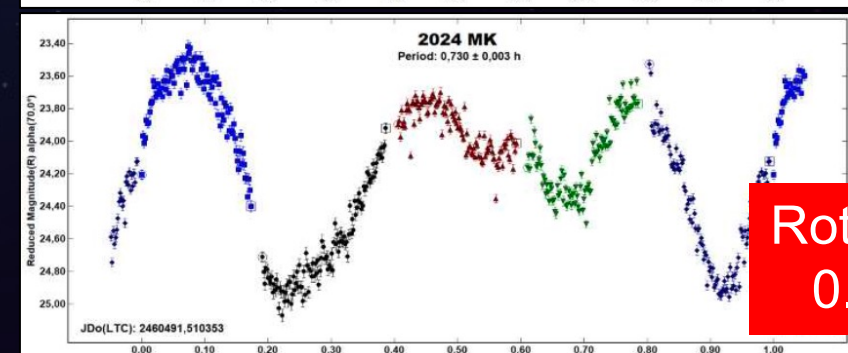
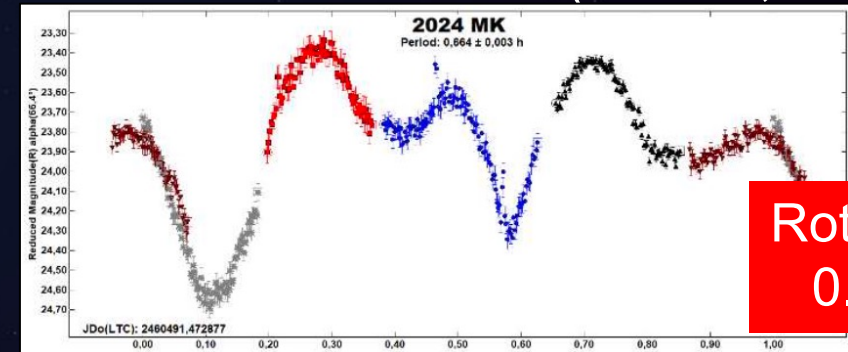
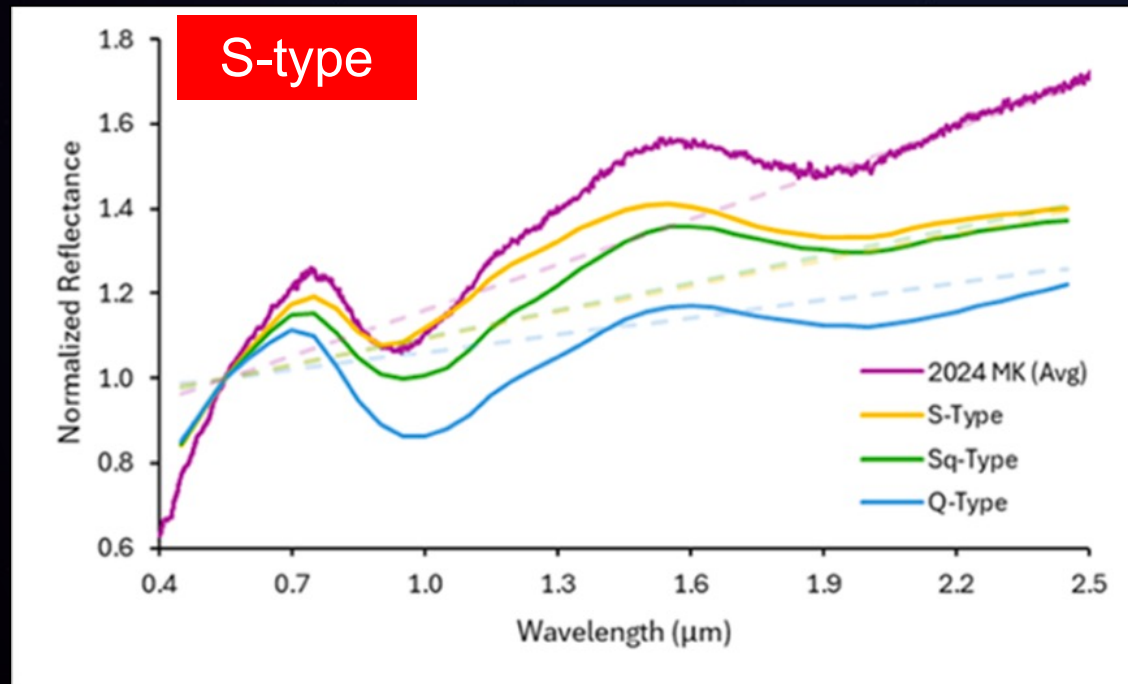


Fig. 39, 40 Vis-NIR reflectance spectral of 2024 MK [McGraw et al., 2024], Light-curve and rotation period of 2024 MK [Sioulas, 2024].

4-4. 2024 MK Observations & Results

【Observations】

- We observed 2 nights; at 29 Jun. and 2 Jul.

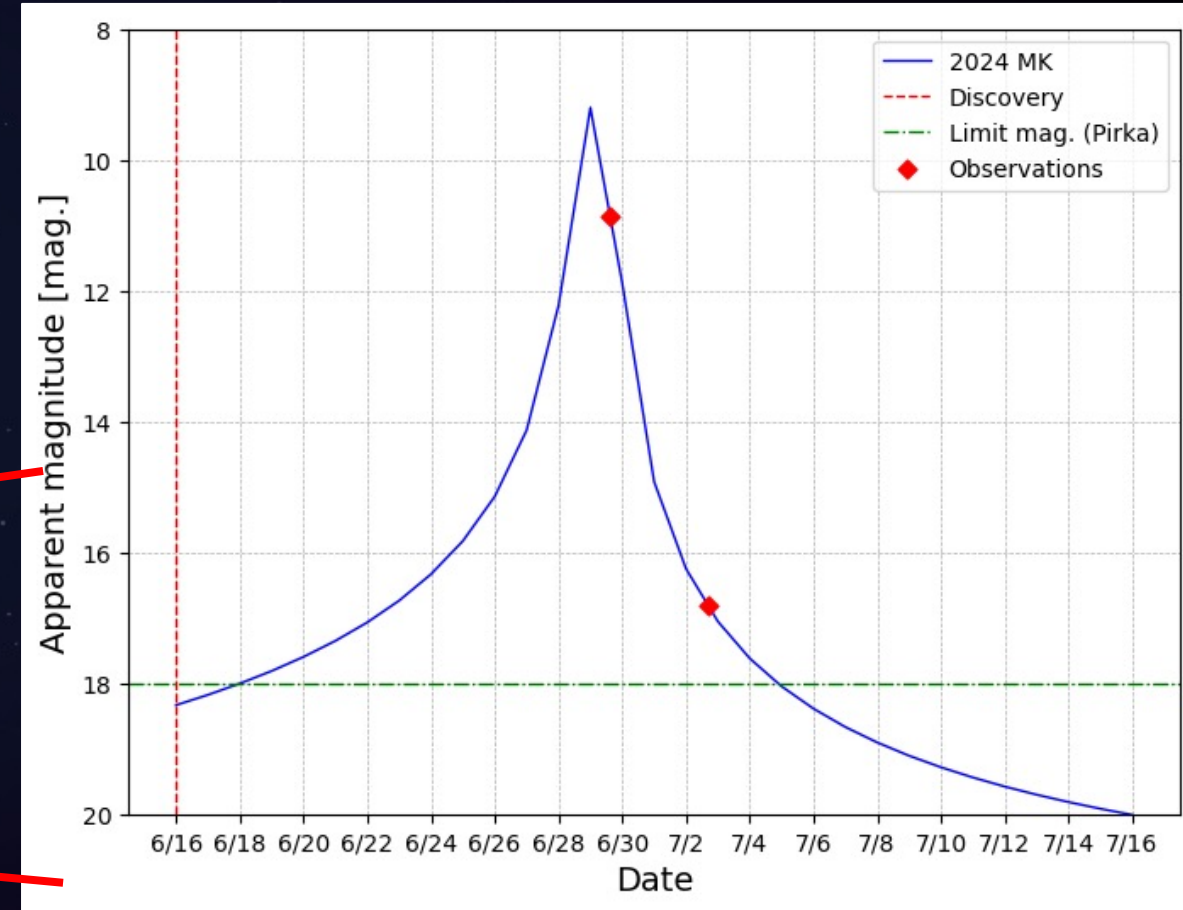
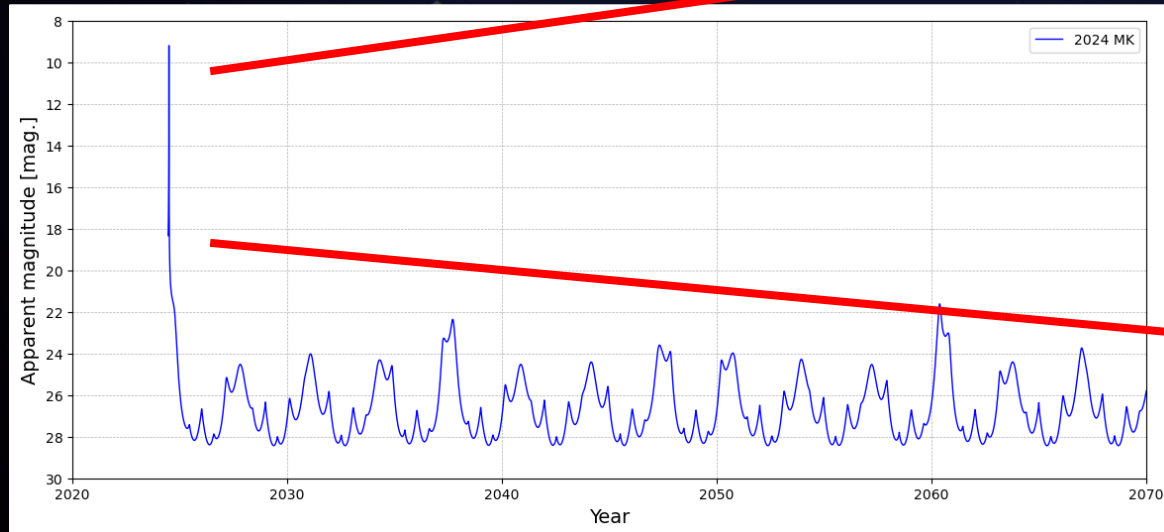
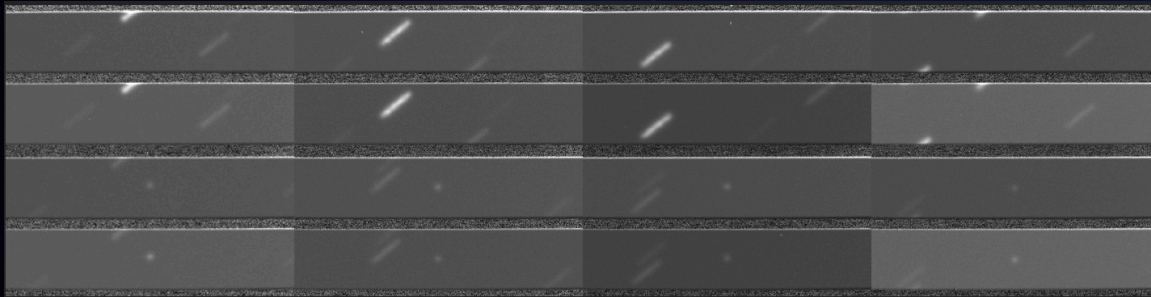


Fig. 41, 42 Point source of 2024 MK at 2 July, Apparent magnitude variation of 2024 MK.

4-4. 2024 MK Observations & Results

Aimed to derive
surface scattering properties

【Observations】

- Polarimetric observations with the Pirka telescope.

Tab. 4 Observation data condition of 2024 MK.

Date	α	Filter	Notes
29 Jun. 2024	26-28°	V, R_c	Closest approach night, S/N ~1500
2 Jul. 2024	101°	V	Max α , S/N ~50, can not be observable after (dark)

Tab. 5 Weather condition around the closest approach.

	6/26	6/27	6/28	6/29	6/30	7/1	7/2
Weather							
V-mag.	15	14	13	9	12	15	17
α	60	58	50-48	15-27	89-90	98-99	100-101
Obs.	×	×	×	○	×	×	○

Over the next
200 years, we
can not observe!

“ichigo-ichie”

4-4. 2024 MK Observations & Results

【Observations】

Tab. 6 Observational conditions of 2024 MK.

Variation is important

Date	UT	Filter	Exp. [s]	V-mag. [mag.]	r [au]	Δ [au]	α [$^{\circ}$]	φ [$^{\circ}$]
29 Jun. 2024	14:46:23-14:48:51	V	30	9.71	1.01840	0.00196	26.351	232.798
29 Jun. 2024	14:50:18-14:52:47	V	30	9.73	1.01840	0.00196	26.765	232.639
29 Jun. 2024	14:55:36-14:58:05	V	30	9.74	1.01839	0.00196	27.386	232.335
29 Jun. 2024	14:58:36-15:01:05	V	30	9.76	1.01839	0.00196	27.696	232.188
29 Jun. 2024	15:06:06-15:08:35	R_c	30	9.78	1.01838	0.00197	28.418	231.816
29 Jun. 2024	15:08:50-15:11:20	R_c	30	9.79	1.01837	0.00197	28.727	231.683
2 Jul. 2024	16:56:48-17:17:19	V	300	16.85	1.01337	0.01683	100.984	249.566

4-4. 2024 MK Observations & Results

【Results】 Liner polarization degree

- Plot of P_r vs α

Tab. 7 Polarization degree of 2024 MK.

Date	Filter	α [$^{\circ}$]	$P_r \pm \sigma P_r$ [$^{\circ}$]	$\theta_p \pm \sigma \theta_p$ [$^{\circ}$]
29 Jun. 2024	V	26.351	1.23 ± 0.12	-26.30 ± 0.51
29 Jun. 2024	V	26.765	1.55 ± 0.20	-15.33 ± 0.67
29 Jun. 2024	V	27.386	1.47 ± 0.15	-24.22 ± 0.51
29 Jun. 2024	V	27.696	1.67 ± 0.16	-22.55 ± 0.49
29 Jun. 2024	R_c	28.418	1.51 ± 0.17	-20.72 ± 0.56
29 Jun. 2024	R_c	28.727	1.43 ± 0.14	-21.44 ± 0.51
2 Jul. 2024	V	100.984	35.15 ± 5.70	-2.48 ± 0.77

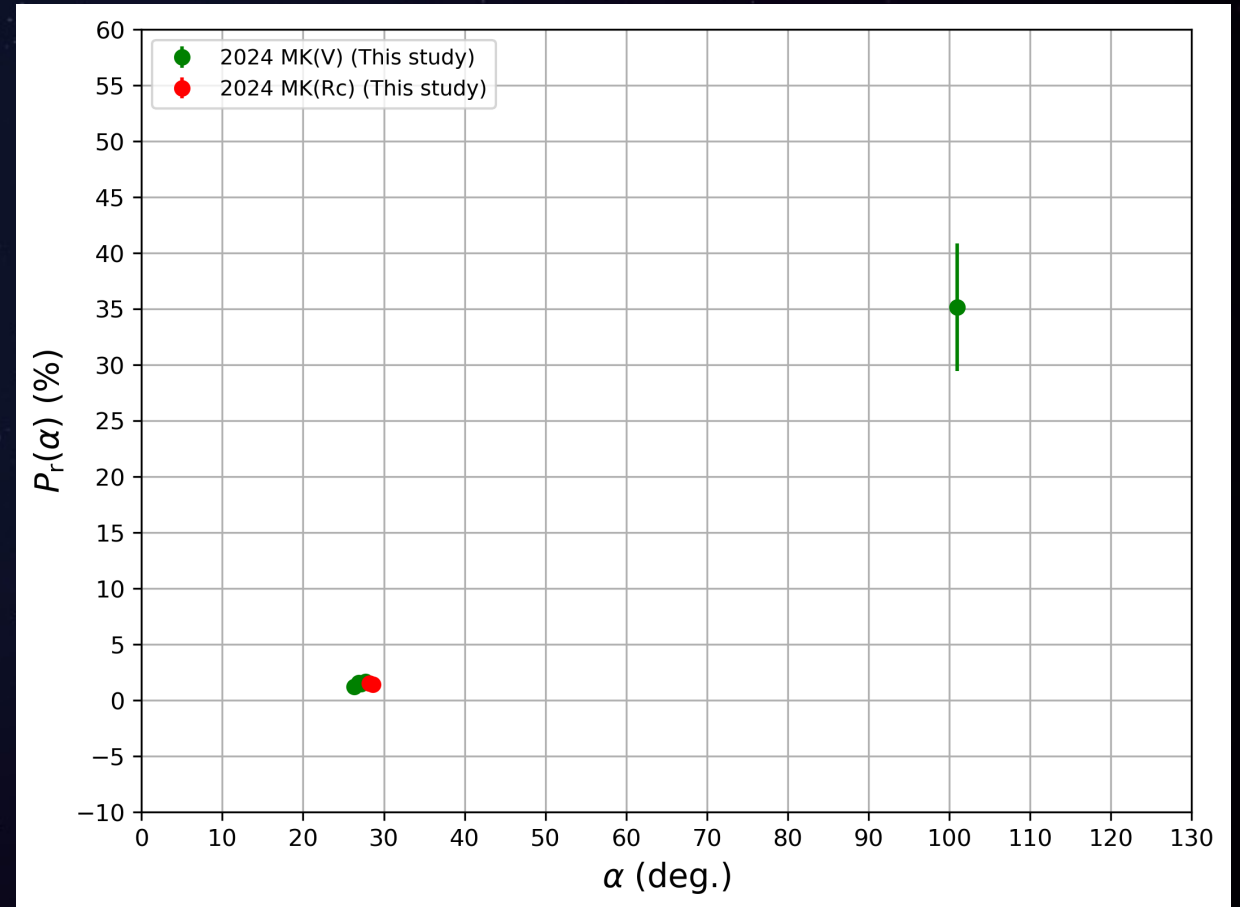


Fig. 43 Polarization degree of 2024 MK.

4-4. 2024 MK Observations & Results

【Results】 Liner polarization degree

- Compared with Liner polarization degree of NEAs.

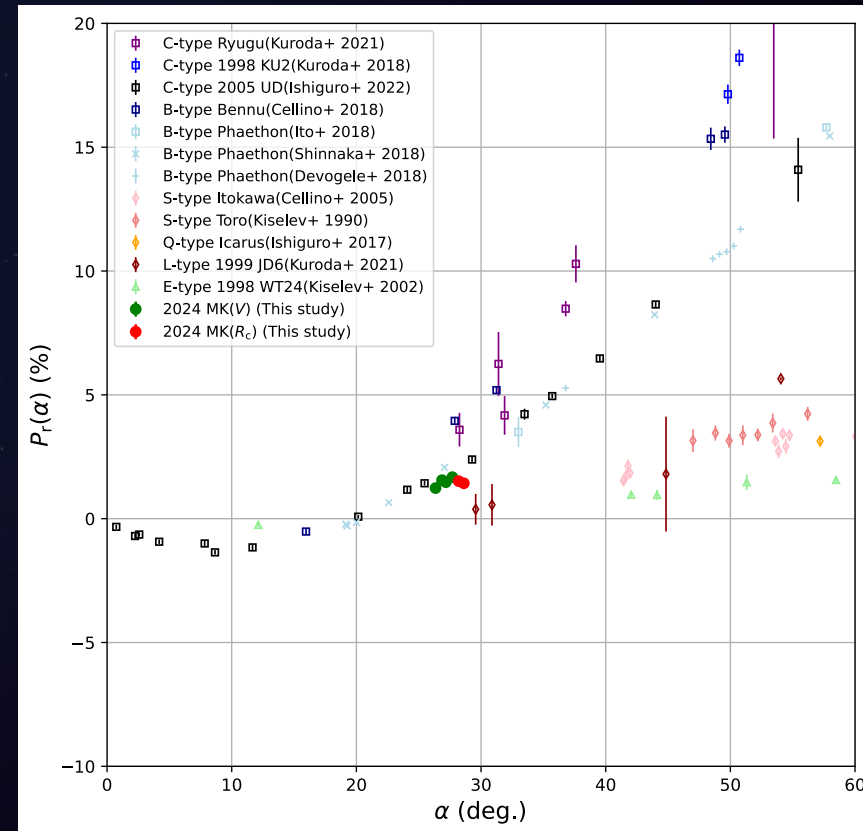
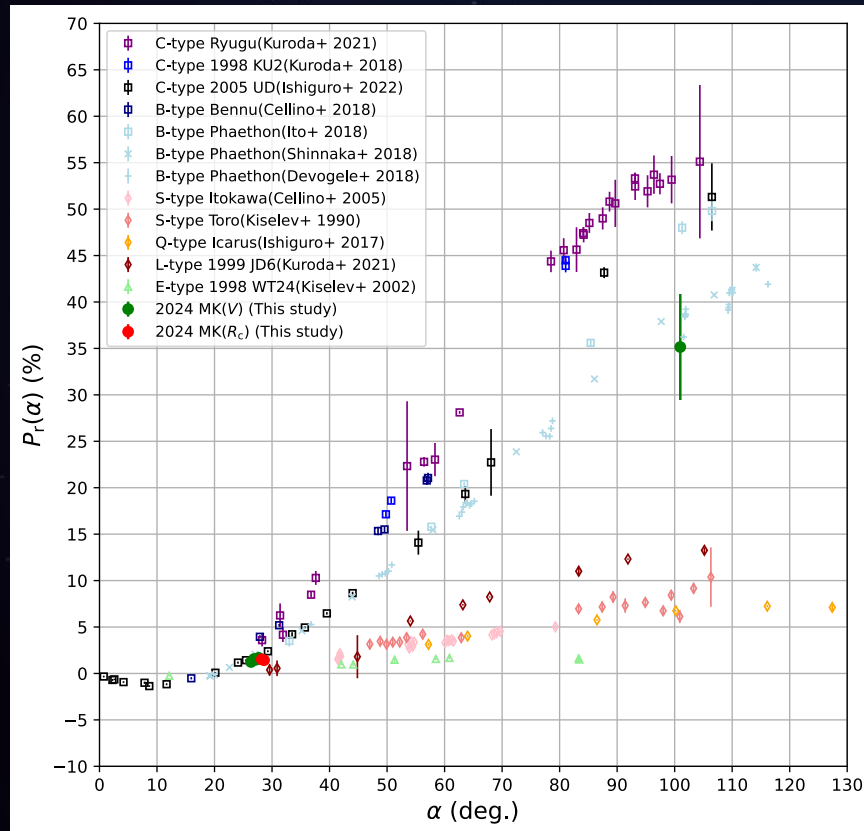


Fig. 45, 46 Polarization degree of 2024 MK and Several previous polarimetric results of NEAs.

4-4. 2024 MK Observations & Results

【Results】 Liner polarization degree

- 2024 MK exhibited high Liner polarization degree.
- It is consistent with C-type.

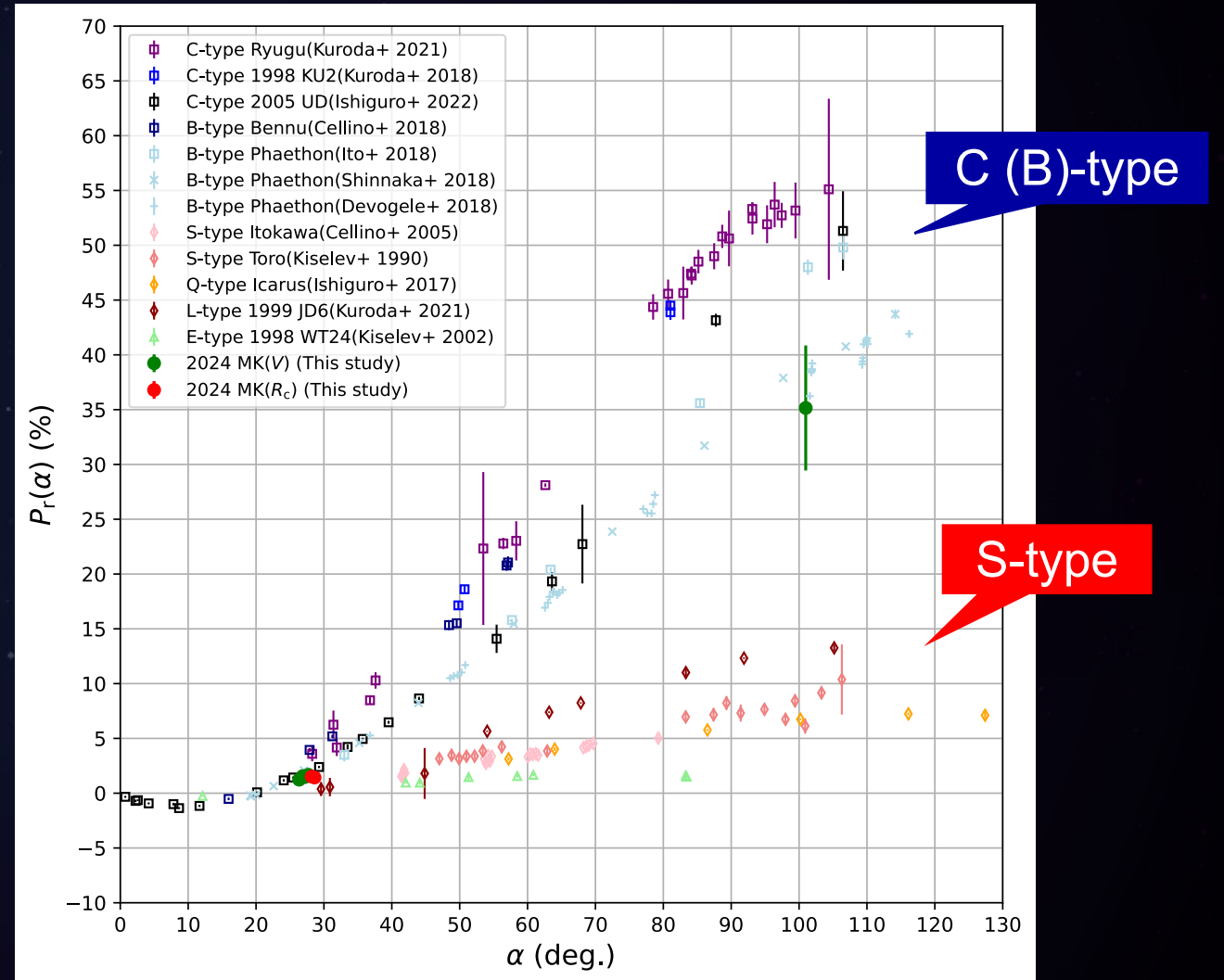


Fig. 47 Polarization degree of 2024 MK and Several previous polarimetric results of NEAs.

4-5. 2024 MK Discussion

【Discussion】

- Spectroscopy derived characteristics of **S-type reflectance spectrum.**
- Polarimetry derived characteristics of **C-type scattering properties.**

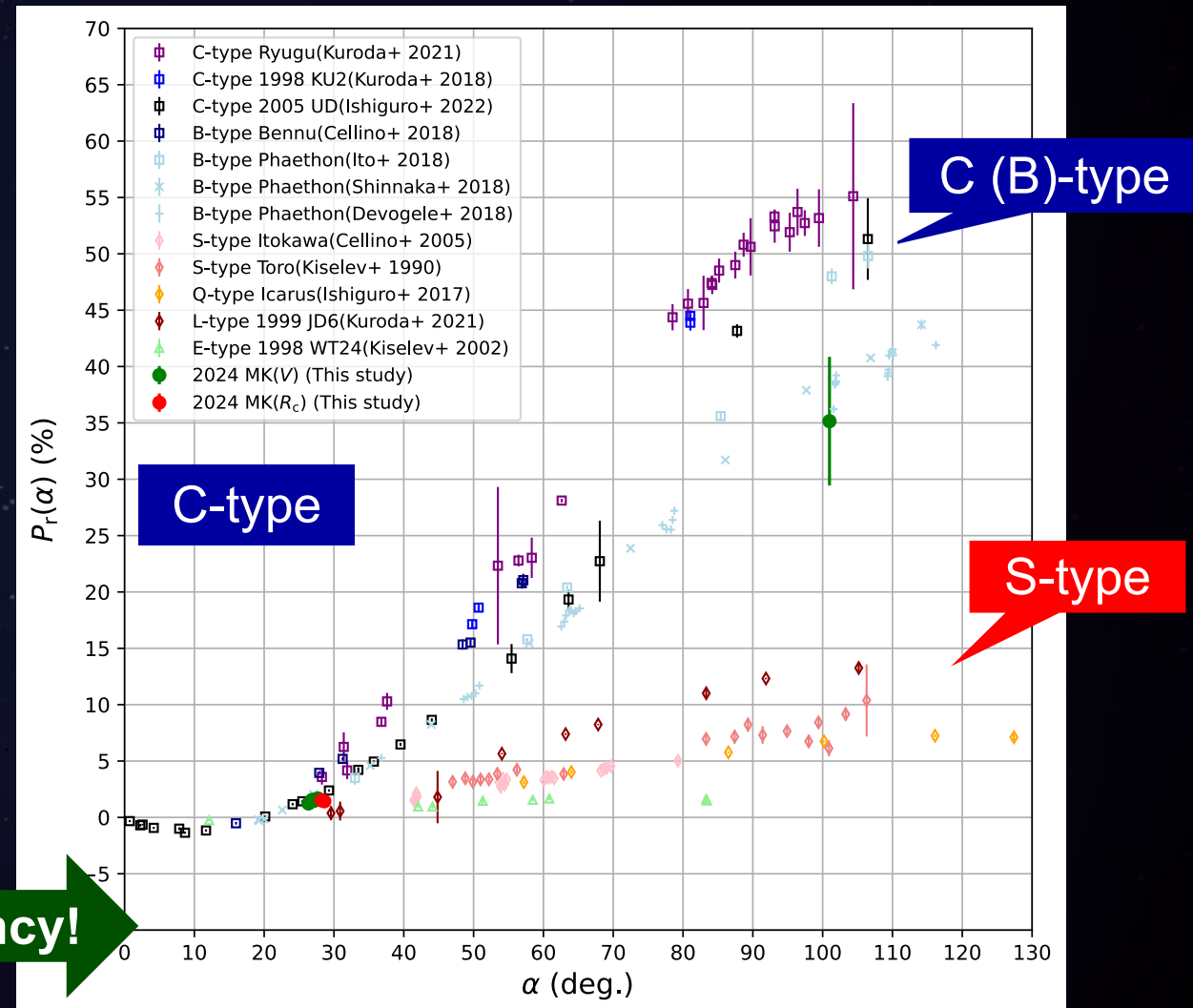
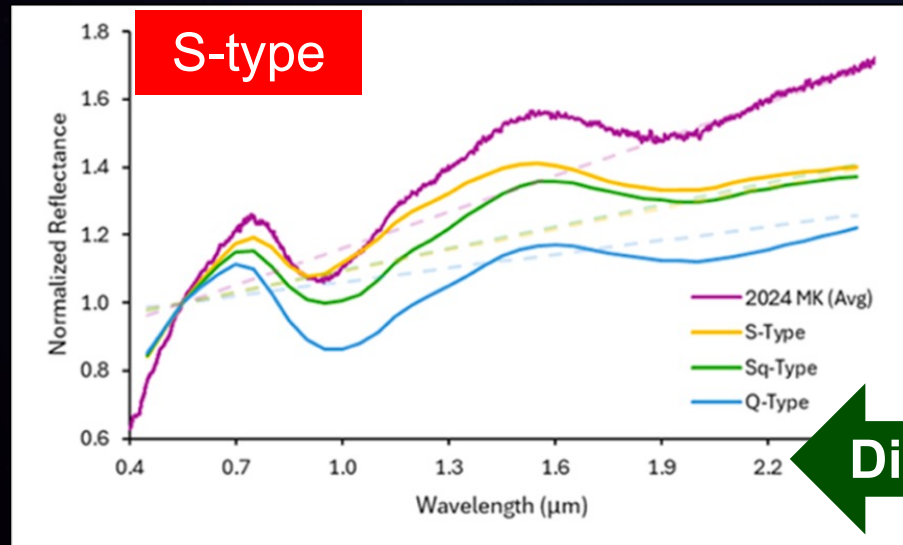


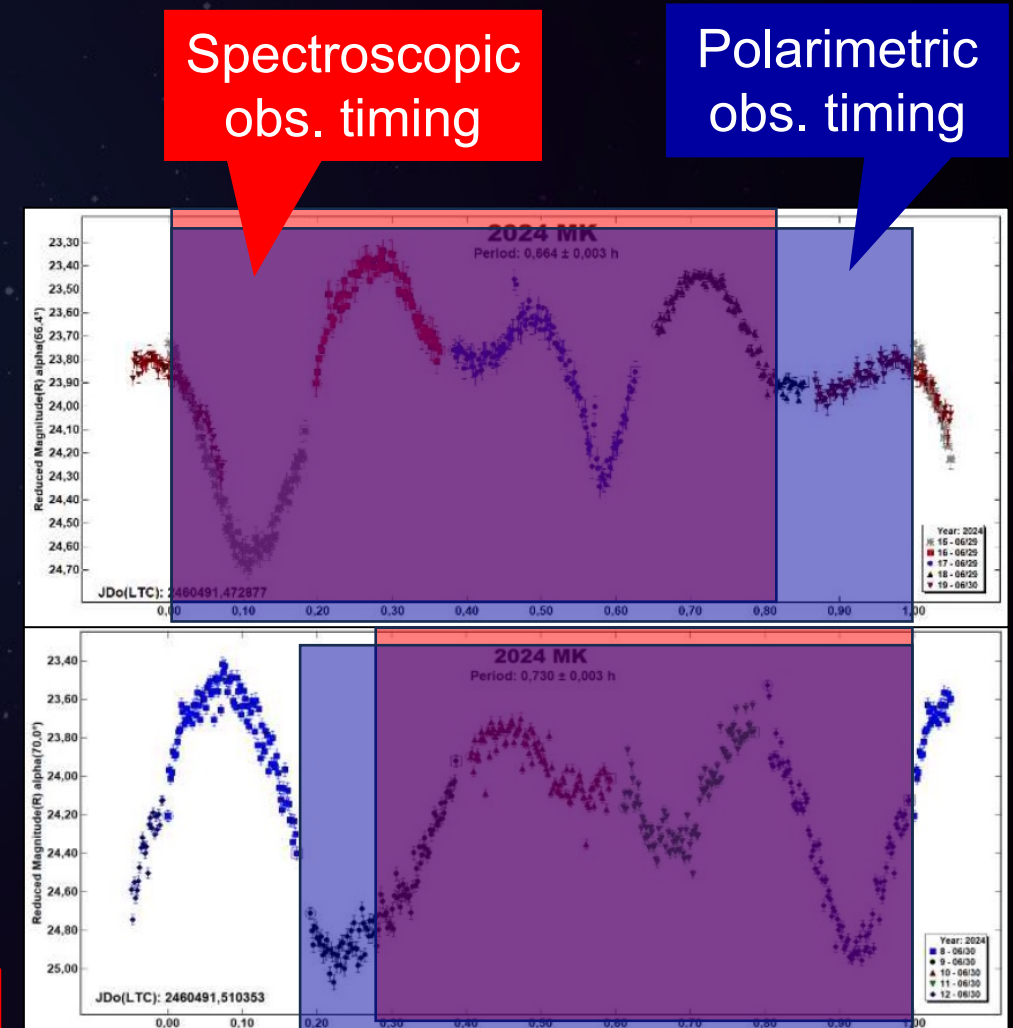
Fig. 39, 47 Vis-NIR reflectance spectral of 2024 MK [McGraw et al., 2024], Polarization degree of 2024 MK and Several previous polarimetric results of NEAs [This study].

4-5. 2024 MK Discussion

【Discussion】

1. Rotation phase (observed different surfaces?)
 - > Differences in the observed surface based on rotation period and observation timing.
 - > We and McGraw et al. (2024) observed the same surface region.
 - > Exhibited both characteristics of S-type reflectance spectrum & C-type scattering properties from same region.

Fig. 48 Light-curve and rotation period of 2024 MK [Sioulas, 2024].

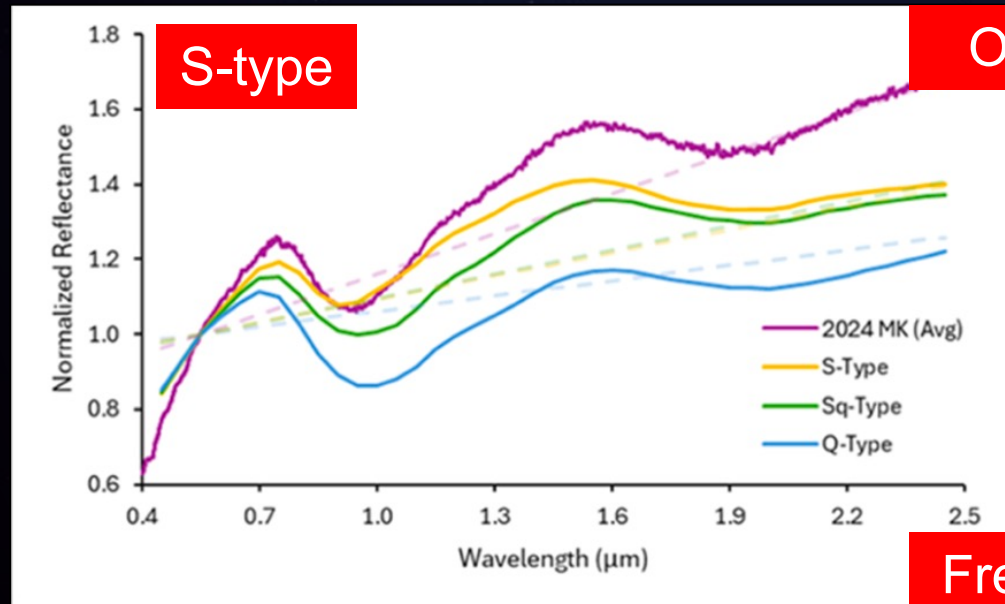


4-5. 2024 MK Discussion

【Discussion】

2. Regolith grain size

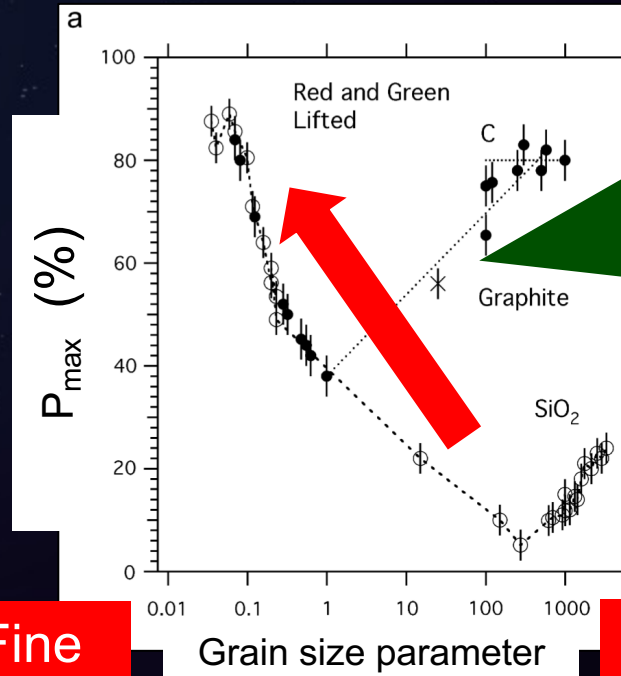
- > May be more space-weathered and/or possess **finer regolith grains** than typical S-type.
- > Presence of fluffy aggregates on the surface increases the Liner polarization degree.
e.g., NEA 1998 KU2 (Kuroda et al., 2018)



Old

Space weathering

Fresh



Old surface
(Space weathered)

Fine grain

High polarization

Fine

Rough

Fig. 39, 30 Vis-NIR reflectance spectral of 2024 MK [McGraw et al., 2024], P_{max} vs Grain size parameter [Hadamcik et al., 2009].

4-5. 2024 MK Discussion

【Discussion】

2. Regolith grain size

-> Assuming $P_{\max} \sim 35.15 \pm 5.70 \%$, estimated grain size: sub-micron – micron size.

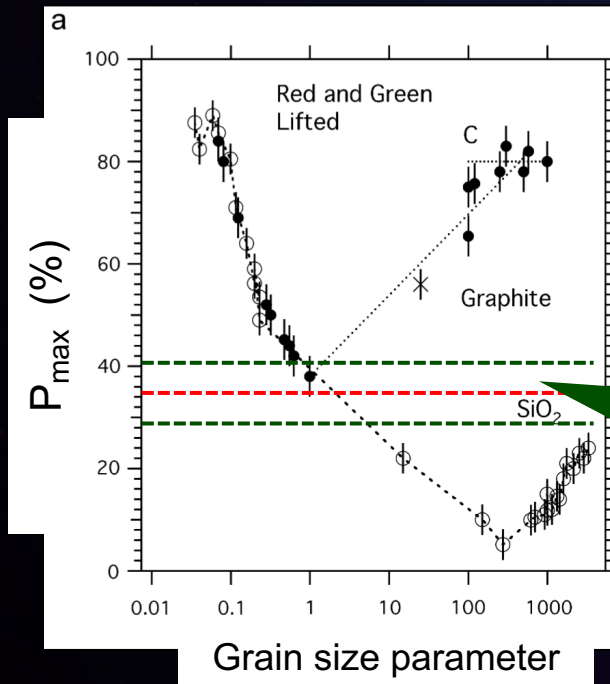
-> In contrast, likely a Monolith and lacks surface grains? (not simple? Rubble-pile?)



Monolith



Rubble-pile



Larger than 300 m are rubble-piles and rotate with a period longer than 2.3 hours

Red line: 35.15 %
Green line: $\pm 5.70 \%$

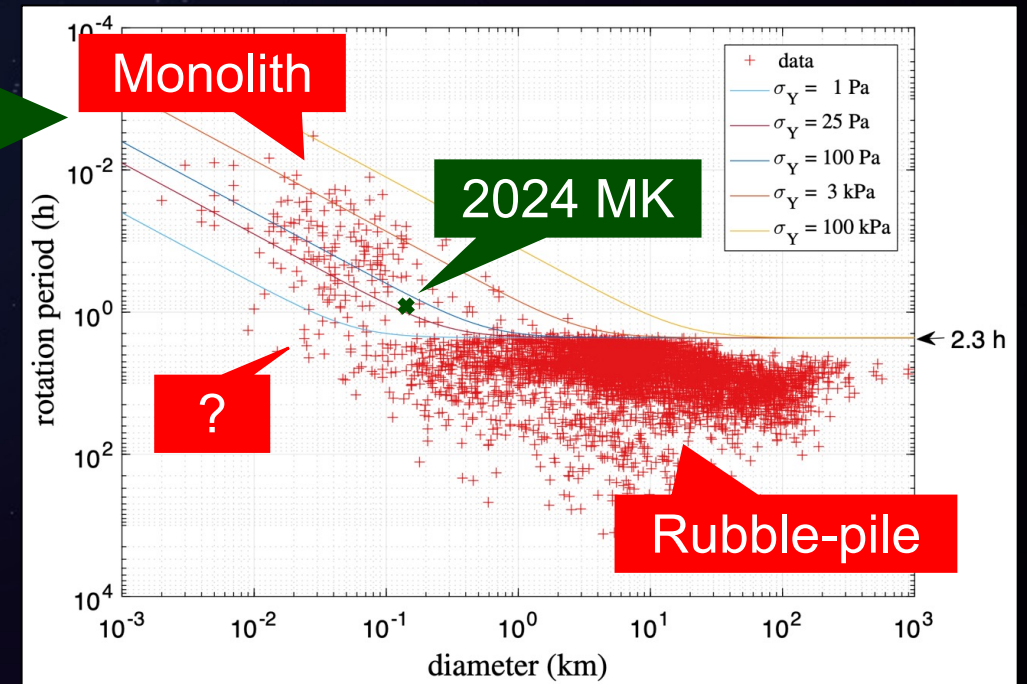


Fig. 30, 48 $P_{\max}$ vs Grain size parameter [Hadamcik et al., 2009], Spin barrier of asteroids [Persson & Biele, 2022].

4-5. 2024 MK Discussion

【Discussion】

3. Porosity

-> High porosity increases the Linear polarization degree, regardless of grain size.
e.g., NEA Phaethon (Ito et al., 2018)



~~1. Rotation phase~~

2. Regolith grain size

3. Porosity

-> Although the overall composition is S-type, its regolith grain size is so fine and/or surface structure is highly porous.

-> We submitted as a letter paper to PASJ at 12 Dec.

5-1. Photometric observations

【Surface color】

- Using broad band filter (e.g., SDSS, Pan-STARRS filter) derived color index of asteroids.
- Color index; g-r, r-i, r-z
g: g-filter mag.
r: r-filter mag.
i: i-filter mag.
z: z-filter mag.

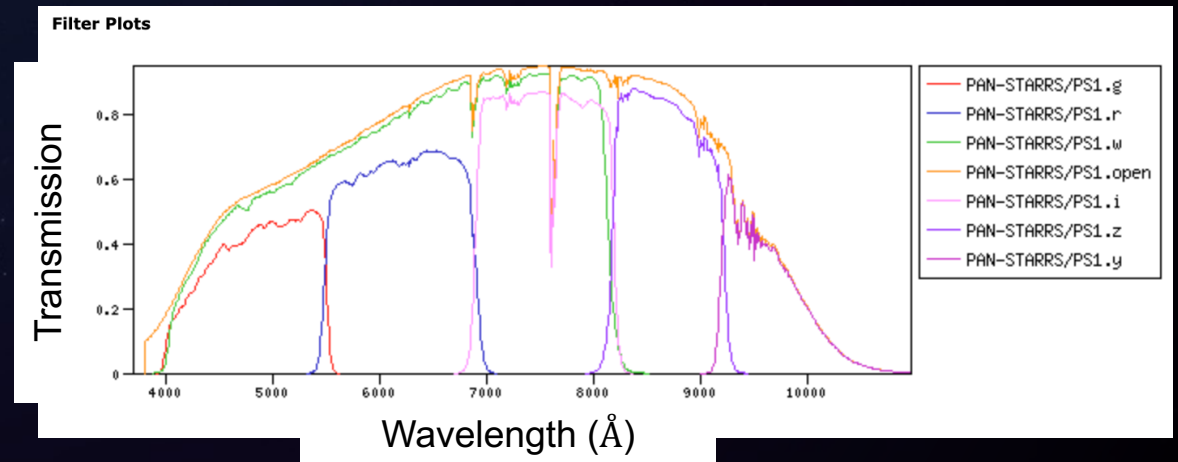
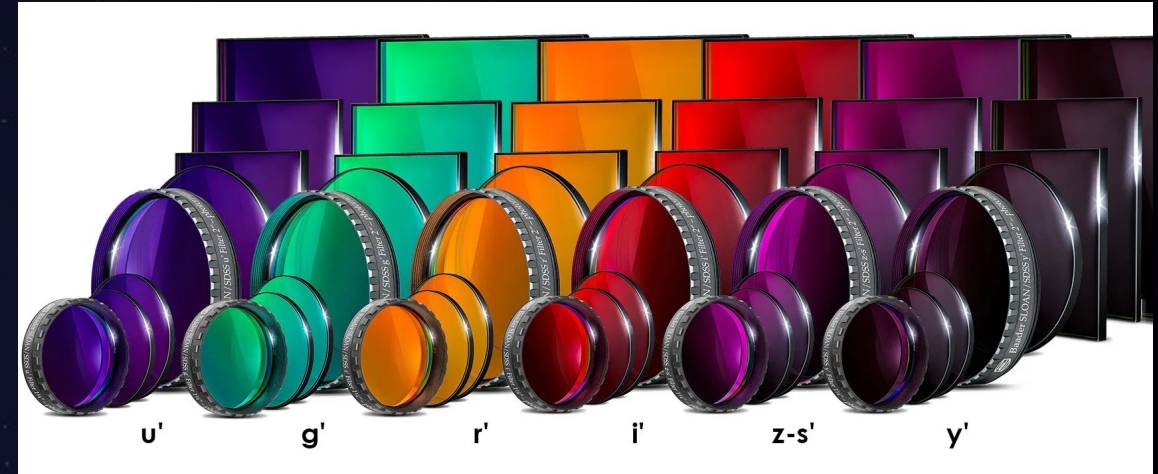


Fig. 49, 50 SDSS system filter, Filter transmission of Pan-STARRS filter.

5-1. Photometric observations

【2 color index figure】

- The asteroid taxonomy can be estimated from color index; g-r, r-i, r-z.

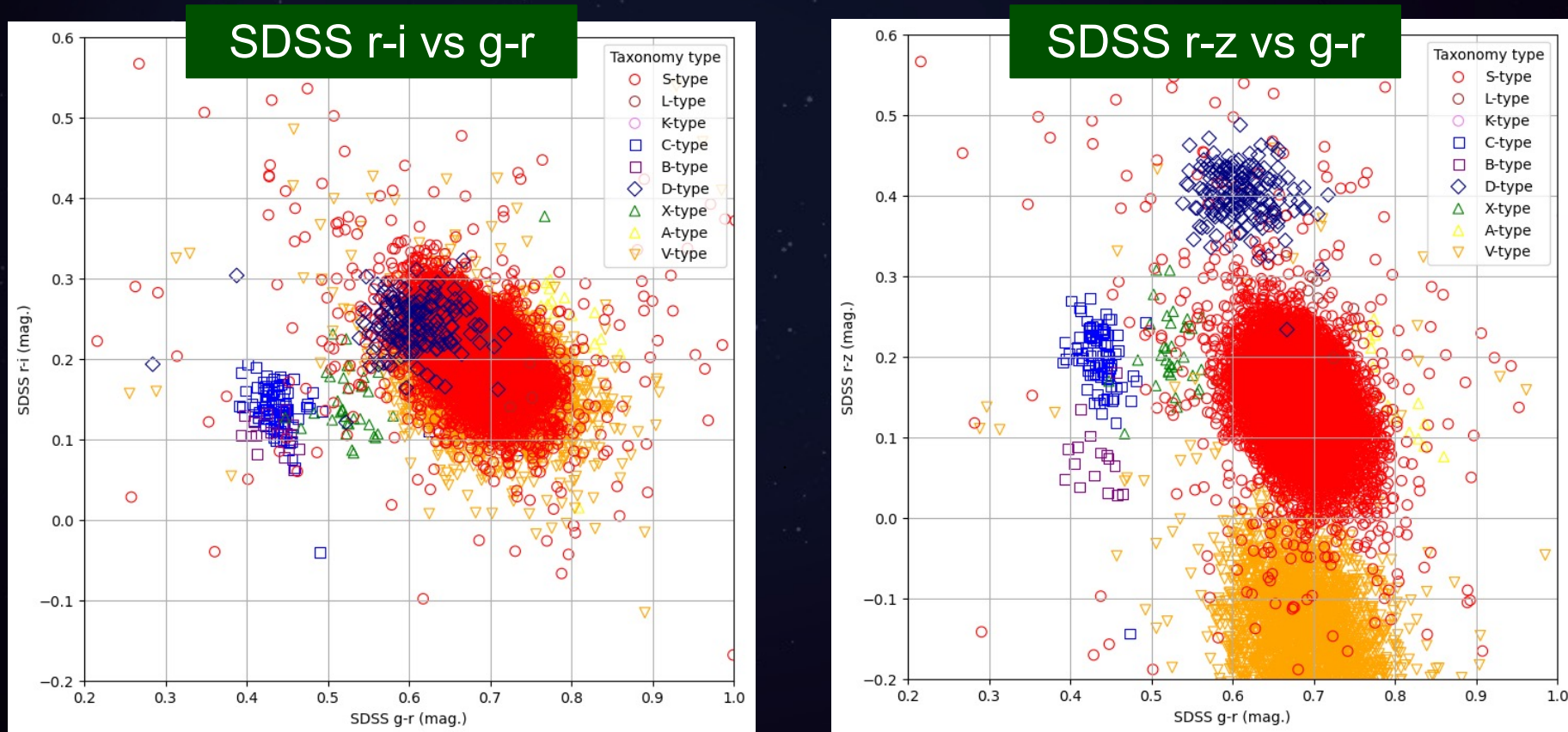
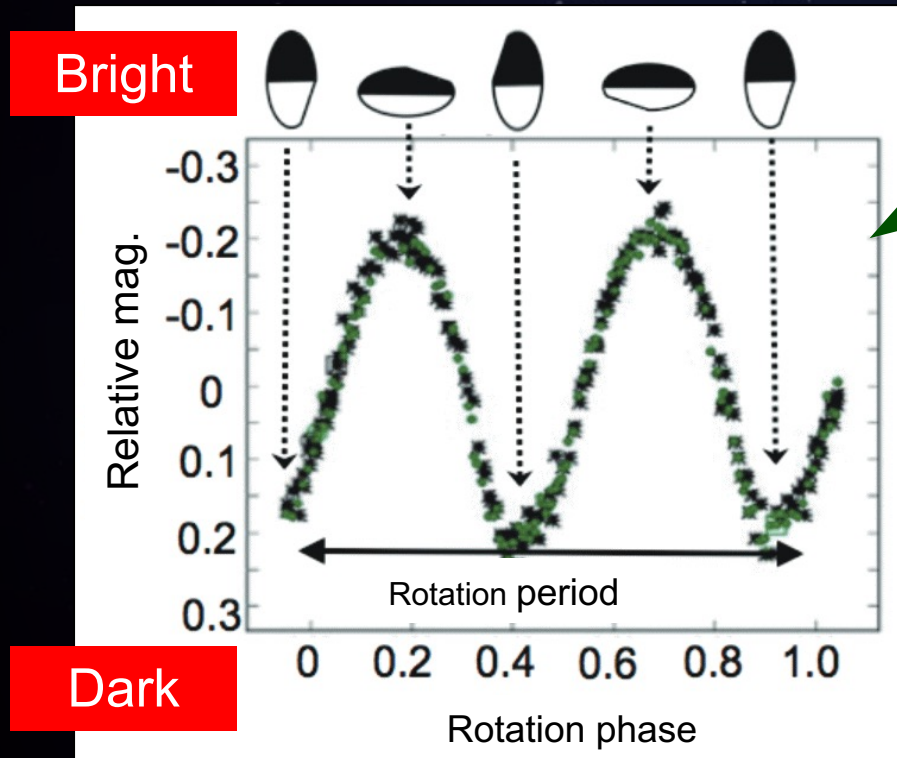


Fig 51. SDSS system 2 color index figure g-r vs r-i & g-r vs r-z of asteroids [Sergeyev & Carry, 2021].

5-1. Photometric observations

【Multi color light-curve】

- Multi color light-curve: Light-curve observations with several broad band filter



(Normal) light-curve;
w/o or 1 filter

Only Rotation period

Multi color light-curve;
w/several filter

The local color of each
rot. phase (rot. period)
can be derived

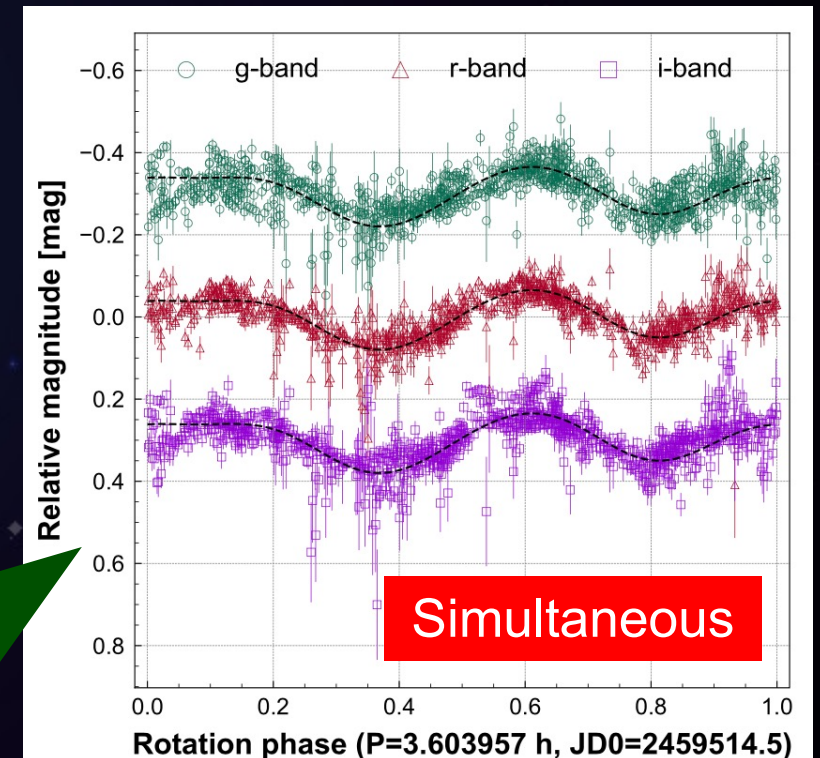


Fig. 52, 53 Rotation period of asteroid, TriCCs color-light curve of NEA [Beniyama et al., 2022].

5-1. Photometric observations

【Color index variation】

- Small color index variation -> Surface homogeneity
Large color index variation -> Surface heterogeneity

This previous study exhibited small color index variation < 0.1 mag.
-> Surface homogeneity

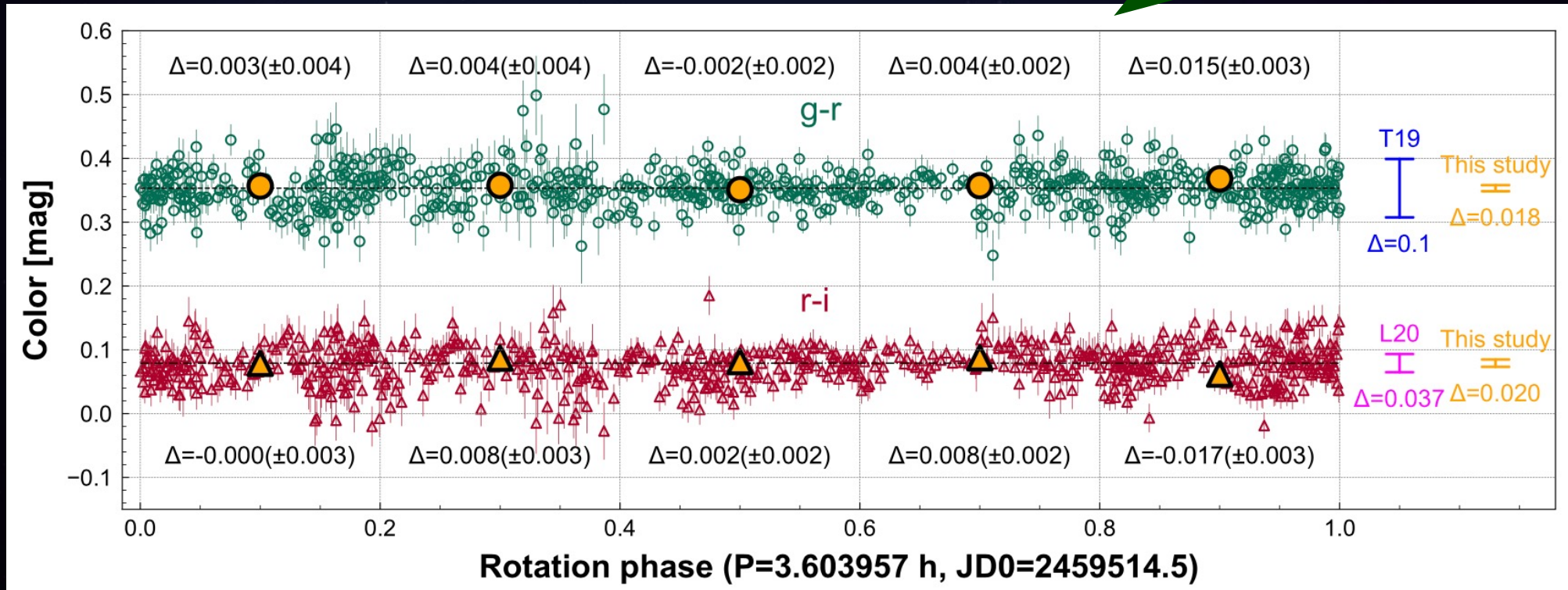


Fig. 54 Color index variation of NEA [Beniyama et al., 2022].

5-2. 3.8 m Seimei Telescope & TriCCs

【Seimei Telescope】

- Location: Okayama Observatory
- Own: Kyoto univ., and open use by NAOJ
The largest optical telescope in East Asia

【TriCCs (Tricolor CMOS Cameras)】

- Imaging mode:
3 (Tri)-colors simultaneous photometry
- FOV: $12.6' \times 7.5'$
- Array format: $2,220 \times 1,360$ pixels
- Filter: Pan-STARRS g, r, i, z

Can conduct Multi color
light-curve photometry

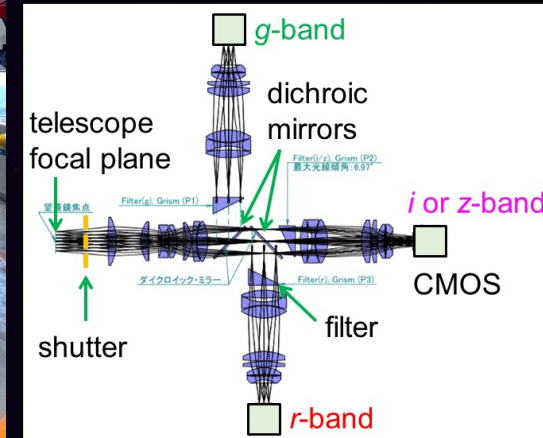
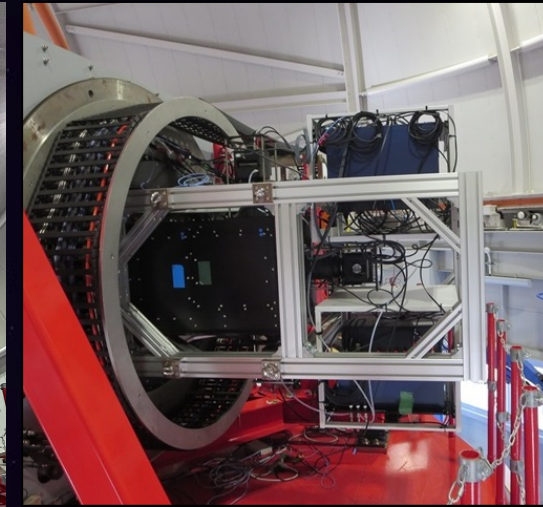


Fig. 55, 56, 57 Seimei Telescope, TriCCs, Imaging mode details of TriCCs.

5-3. Torifune Information

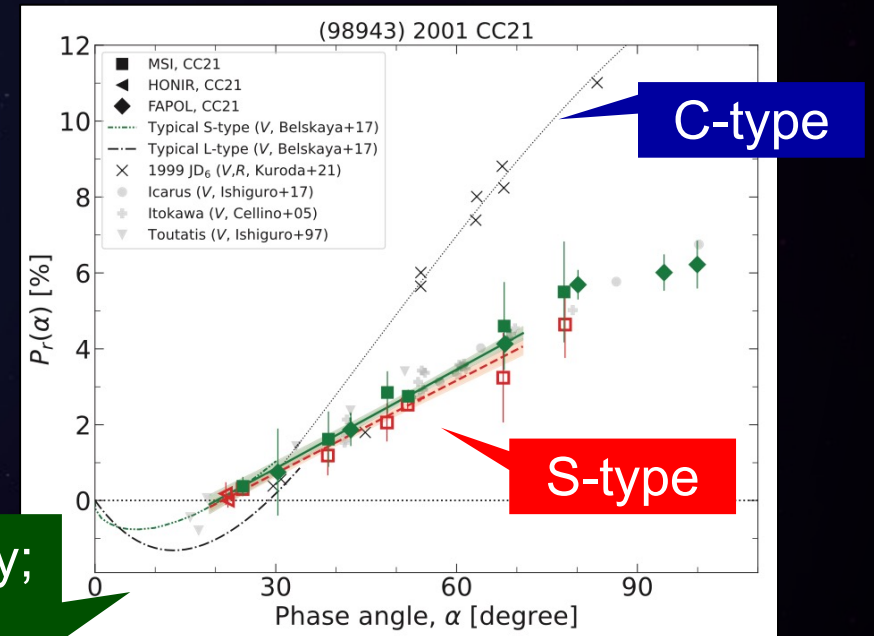
【Information】

- Hayabusa2# next target NEA (PHA), named by JAXA last year (previously 2001 CC21).
- Flyby in 5 July 2026 (next year!)
- Similar to (Hayabusa targets asteroid) Itokawa?

Tab. 8 Comparison of Itokawa and Torifune.

	Itokawa	Torifune
Taxonomy	S-type	S (Sq-) type
Size, Shape	535 × 294 × 209 m	~500 m
Surface	Rubble pile	?

Polarimetry;
S-type



【Other studies】

- Polarimetric observations derived **S-type** Liner polarization degree (Geem et al., 2023).

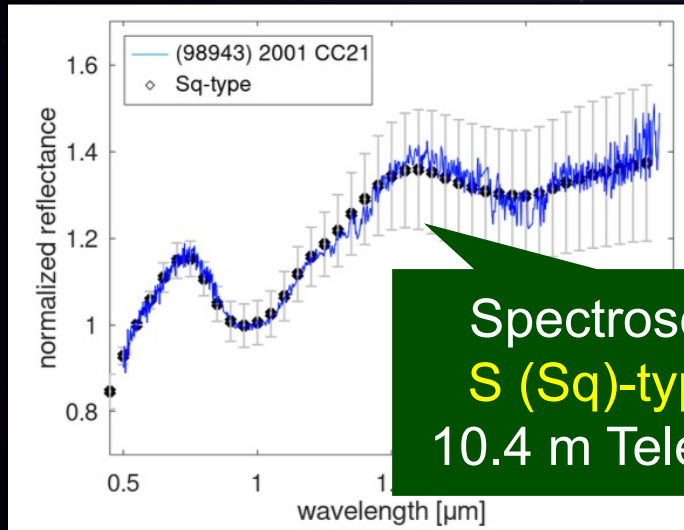
Fig. 58 Polarization curve of Torifune [Geem et al., 2023].

5-3. Torifune Information

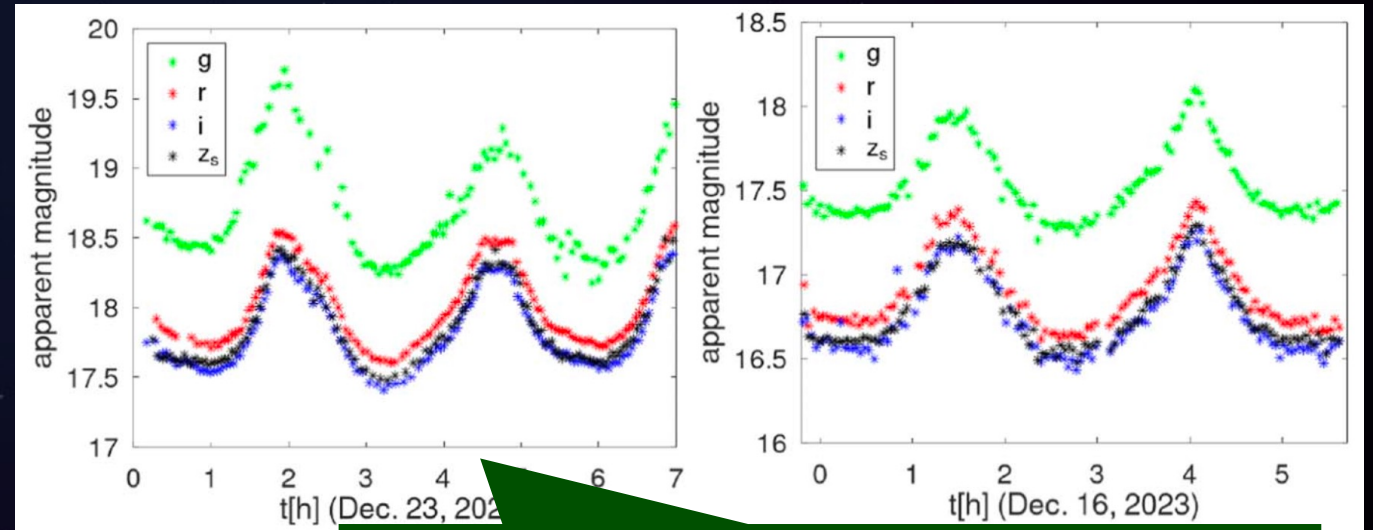
Sq-type asteroids are object that have experienced moderate space weathering

【Other studies】

- Spectroscopic observations derived **S (Sq)-type** reflectance spectrum (Popescu et al., 2025).
- Photometric observations derived **S-type** color index (Mean $g-r = 0.563$, $r-i = 0.178$, $r-z = 0.098$) and large-scale **surface homogeneity** (Popescu et al., 2025).



Spectroscopy;
S (Sq)-type by
10.4 m Telescope



Photometry; **S-type** and **homogeneity**
by 1.5 m Telescope (S/N > 50)

Fig. 59, 60 Reflectance spectrum of Torifune, color light-curve of Torifune [Popescu et al., 2025].

5-3. Torifune Information

【Other studies】

- Light-curve observations derived rotation period of **5.02 hours** (Popescu et al., 2025, Fatka et al., 2025, Fornasier et al., 2024b) and 3D model (Popescu et al., 2025).

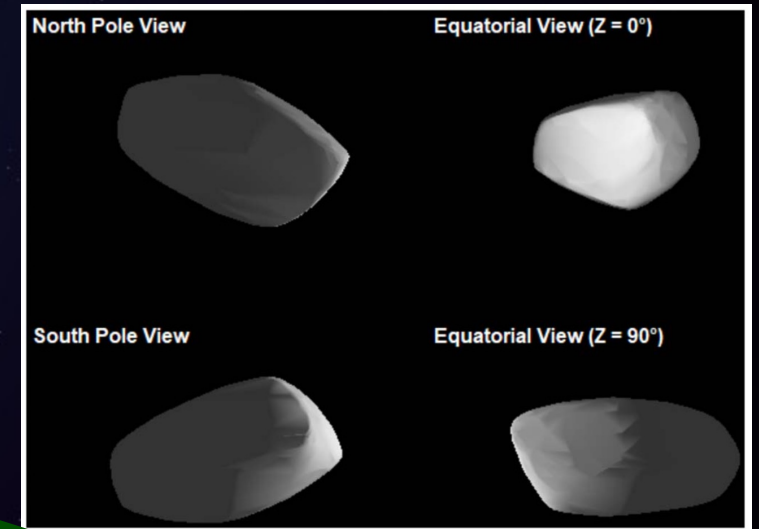
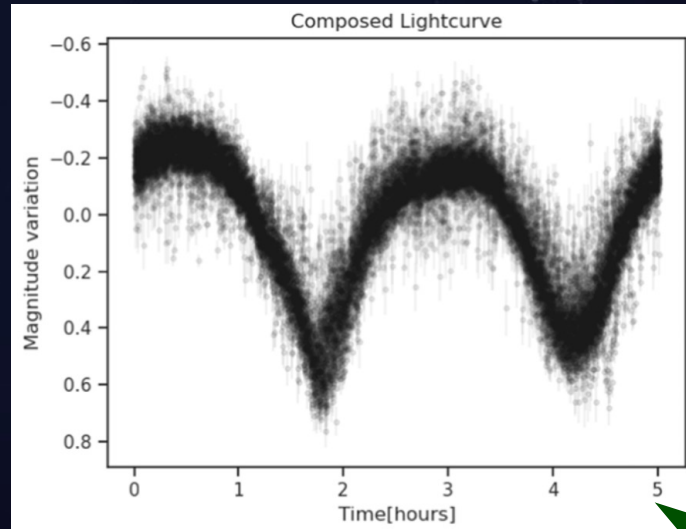
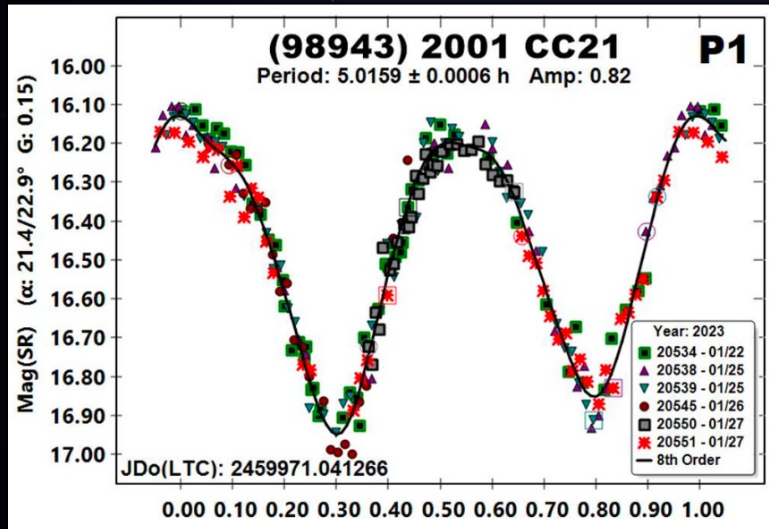


Fig. 61, 62, 63 Light-curve of Torifune [Popescu et al., 2025, Fatka et al., 2025, Fornasier et al., 2024b], 3D model of Torifune [Popescu et al., 2025].

Photometry (Light-curve);
5.02 hours

5-4. Torifune Observations & Results

【Observations】

- We submitted 2 proposals and were accepted by NAOJ.
- We were allocated 3 nights, and 1 additional night of DDT.

Tab. 9 Observation data condition of Torifune.

Date	α	Filter	Exp. [s]	Obs. time	Notes
27 Nov. 2024	5-6°	<i>g, r, i</i>	120	~20 minute	Cloudy
28 Nov. 2024	6-7°	<i>g, r, i</i>	120	~3 hours	Sunny -> Cloudy
22 Oct. 2025	20°	<i>g, r, i</i> & <i>g, r, z</i>	60	~2 hours	Cloudy -> Sunny
6 Nov. 2025	25°	<i>g, r, i</i> & <i>g, r, z</i>	120, 180	Over 5 hours	DDT (Remote) Full moon

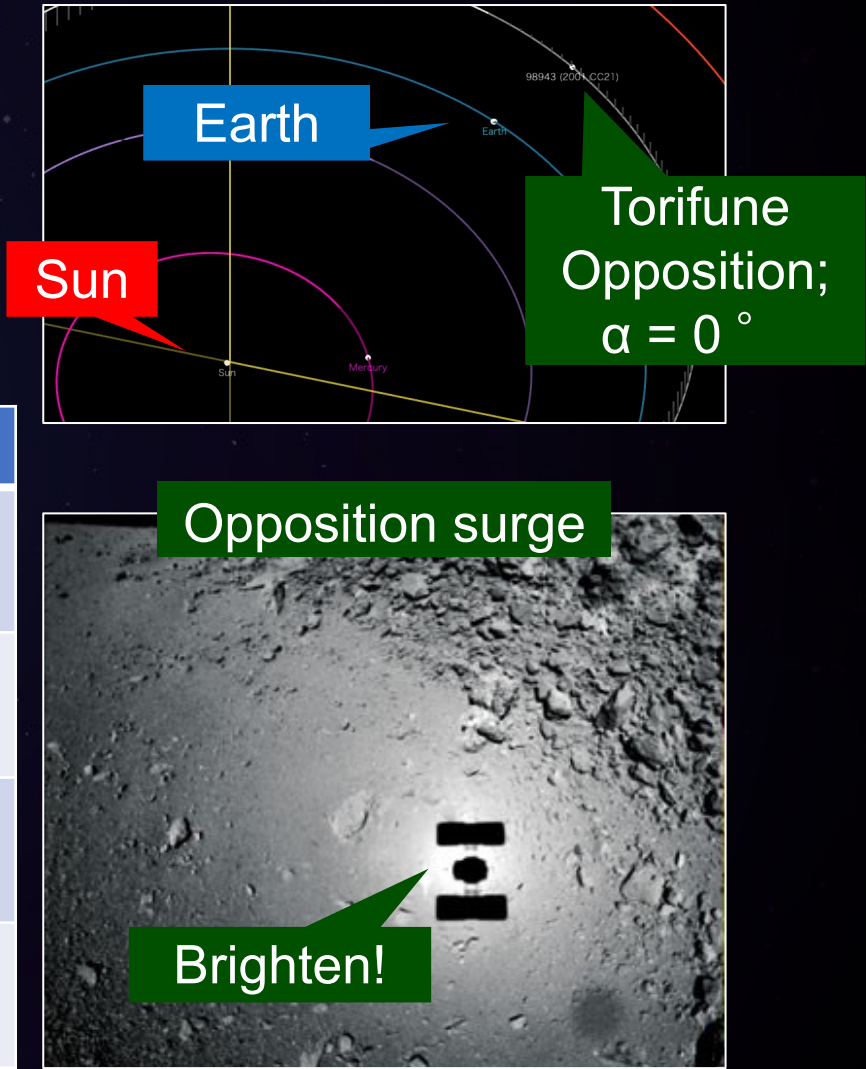


Fig. 64, 65 Torifune's position at 11/27-28, Opposition surge of Itokawa and Hayabusa.

5-4. Torifune Observations & Results

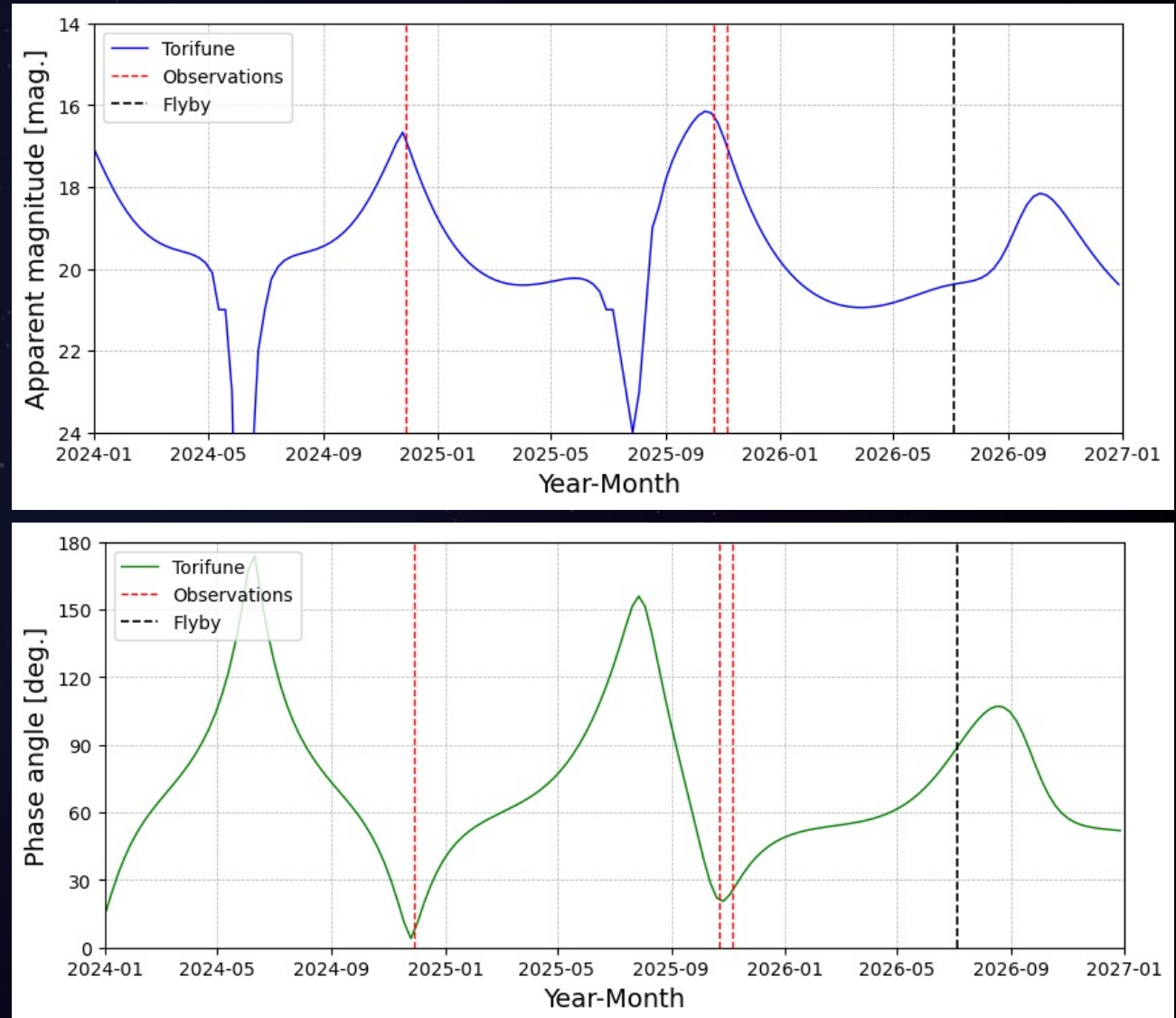
【Observations】

- (3 color simultaneous) Photometric observations with the Seimei telescope.

Aimed to derive **surface color**

- Using the opposition surge (brighten!).

Fig. 66, 67 Apparent magnitude variation, Phase angle variation of Torifune.



5-4. Torifune Observations & Results

【Results】 Multi color light-curve (2024.11.28)

- Great multi color light-curve!
- Each point is 12 s exp. time data (Non-stacked, S/N = 30-60).
-> High time resolution!
- Data gaps were due to cloudy conditions.

Non-stacked multi color light-curve
on 2024.11.28 (Rot. period)

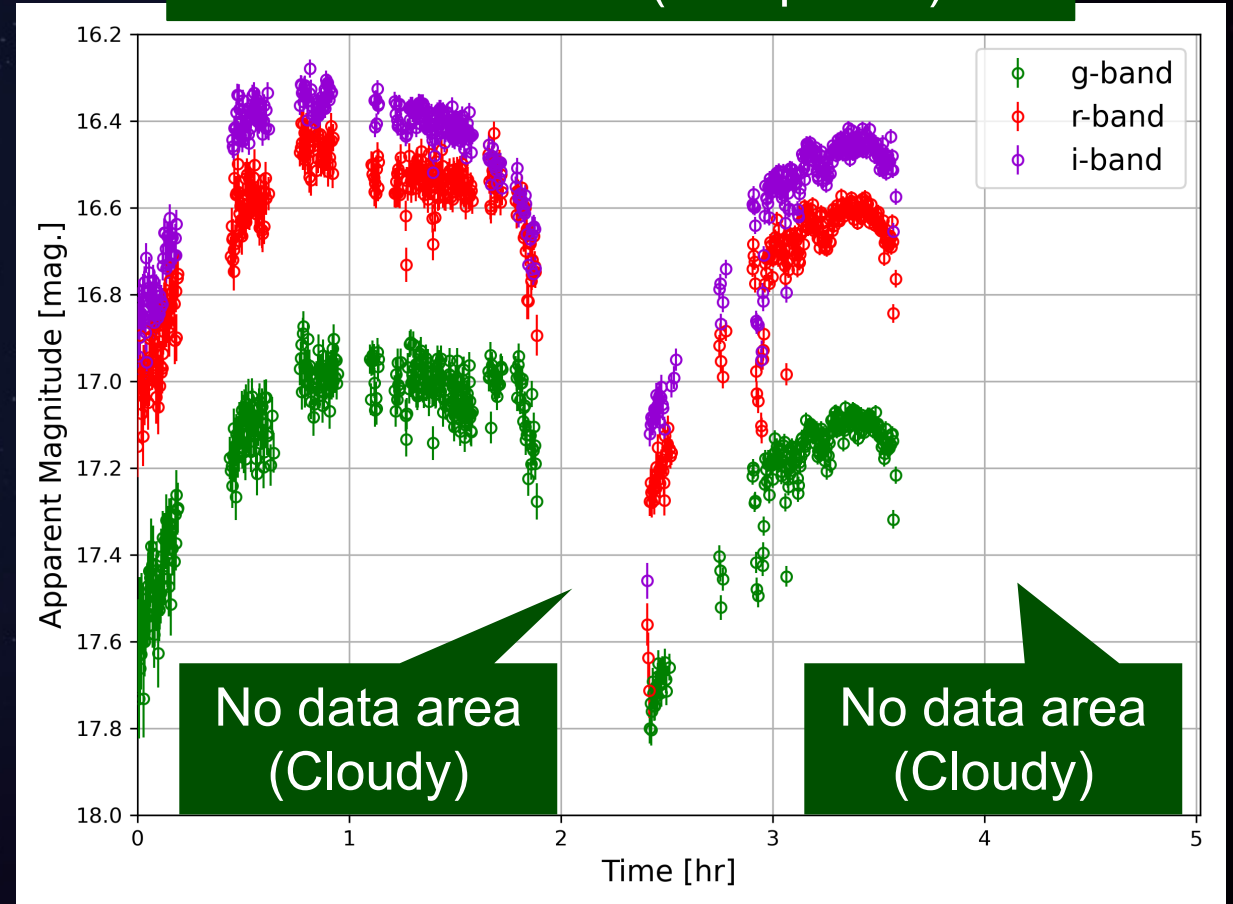


Fig. 68 Non-stacked multi color light-curve on 11.28 [This study].

5-4. Torifune Observations & Results

【Results】 Multi color light-curve (2024.11.28)

- Each point is 120 s exp. time data (Stacked, S/N = 100-150).
-> High photometric accuracy!

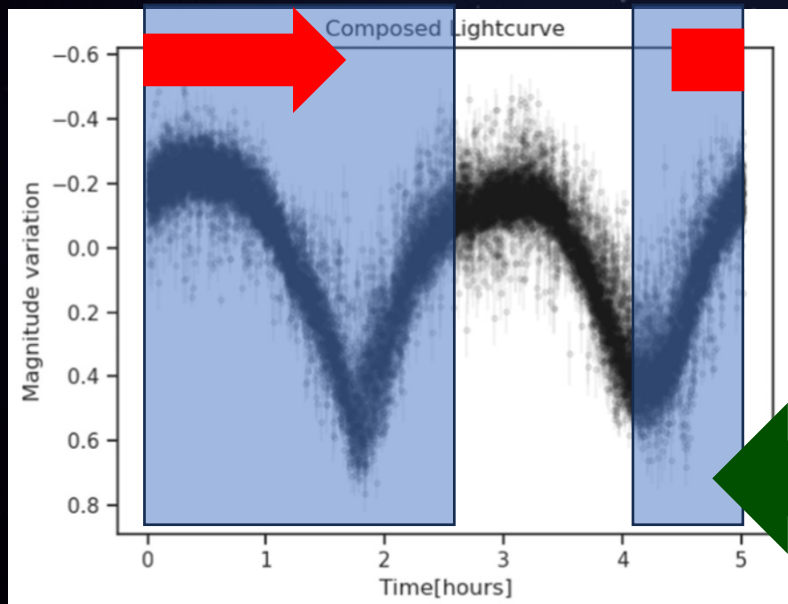


Fig. 62 Light-curve of Torifune [Popescu et al., 2025].

Stacked multi color light-curve
on 2024.11.28 (Rot. period; 5.02 h)

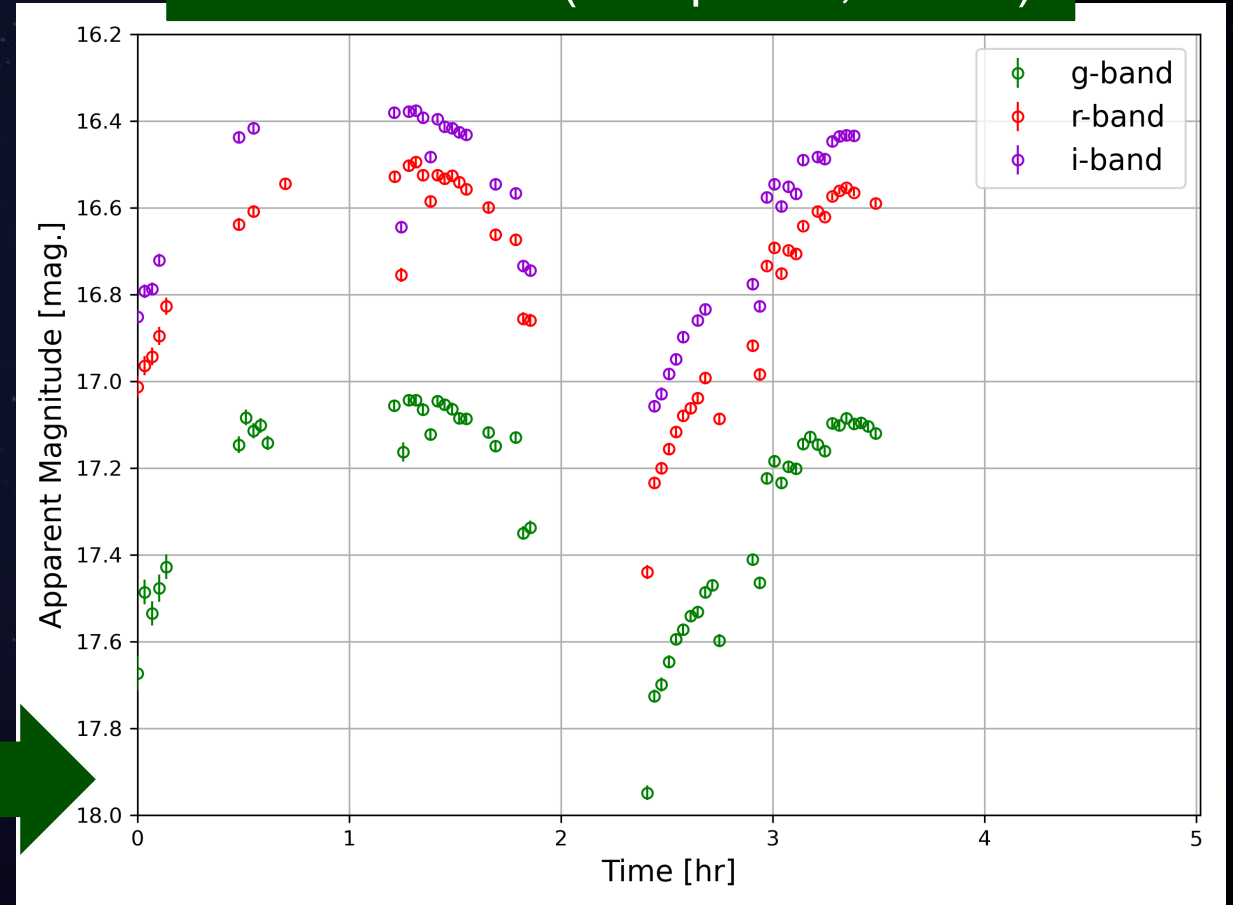


Fig. 69 Stacked multi color light-curve on 11.28 [This study].

5-4. Torifune Observations & Results

【Results】 Multi color light-curve (2025.10.22)

- Each point is 60 s exp. time data (Stacked, S/N ~ 100).
- area 1
 - > Sky condition was bad...
 - > Color index is unstable
- area 2
 - > Sky condition was good
 - > Color index is stable
 - > Used for mean color calculation

Stacked multi color light-curve
on 2025.10.22 (Rot. period; 5.02 h)

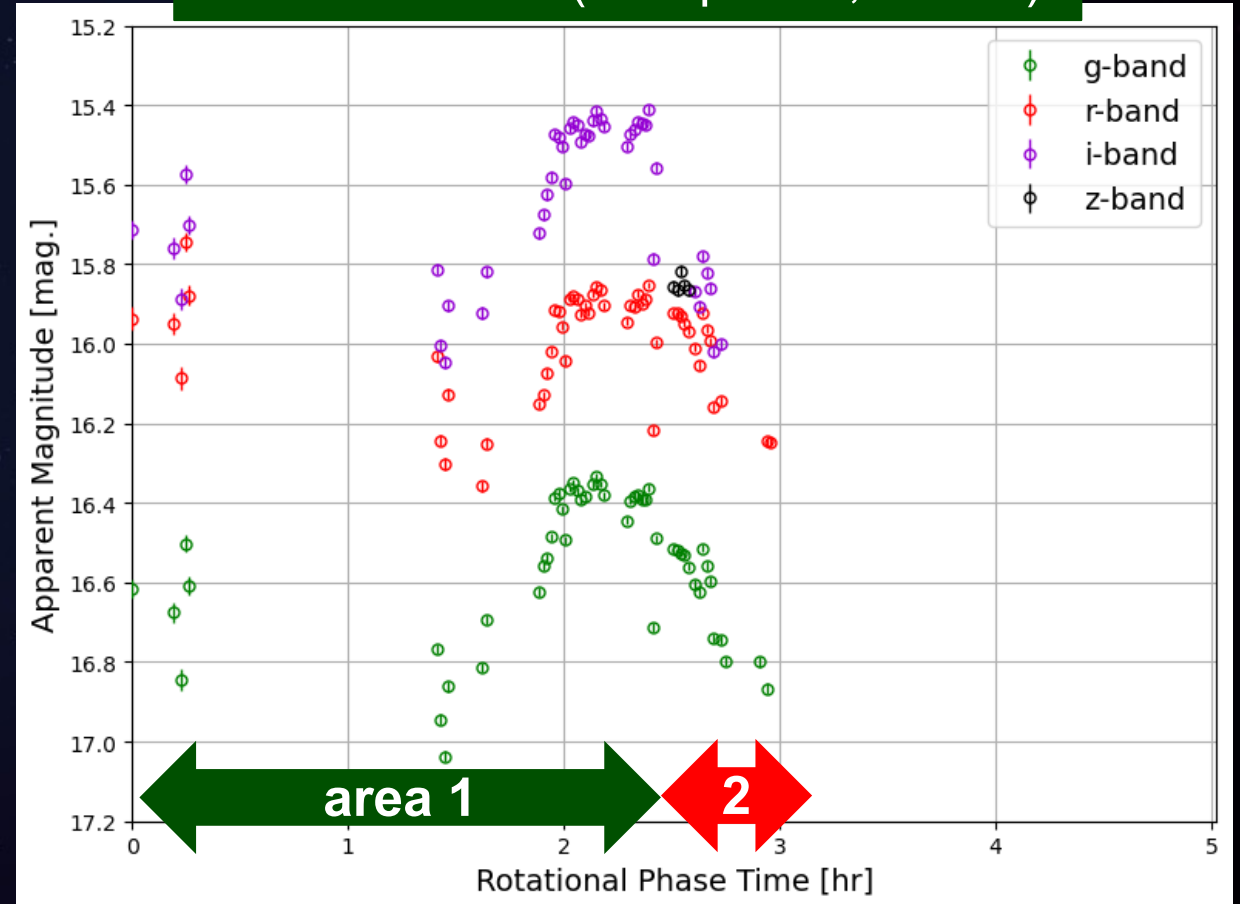


Fig. 70 Stacked multi color light-curve on 10.22 [This study].

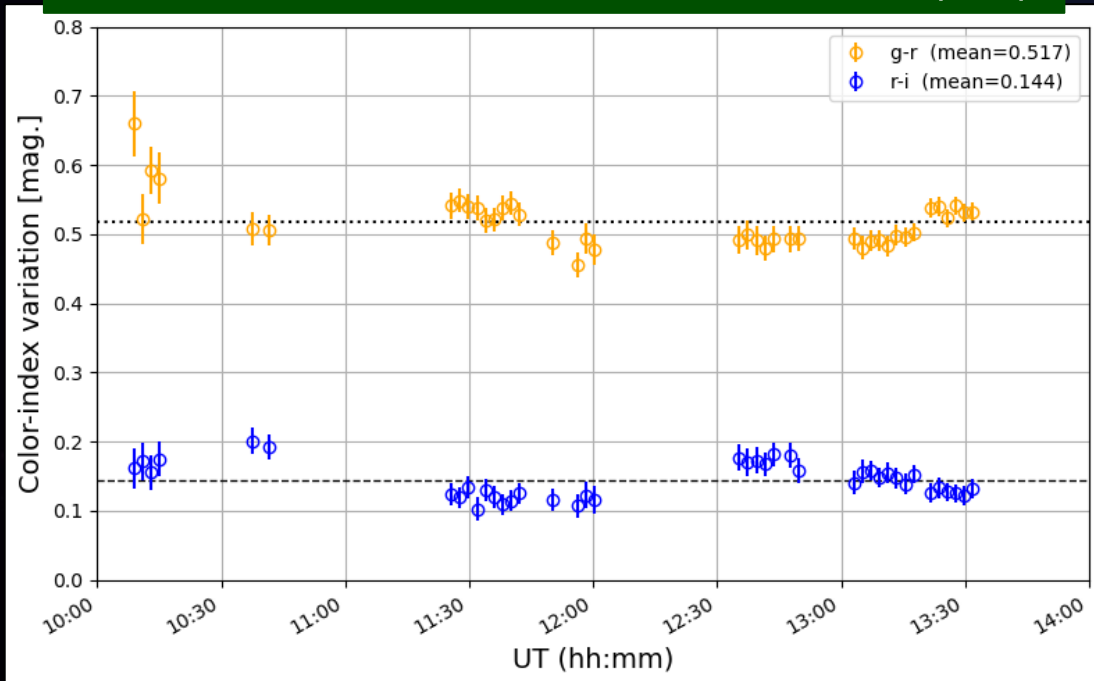
5-4. Torifune Observations & Results

【Results】 Color index variation (2024.11.28 & 2025.10.22)

- We confirmed small color index (g-r, r-i, r-z) variation.
-> Large-scale surface homogeneity

The following mean values are calculated using only the area 2 data

Color index variation on 2024.11.28 (UT)



Color index variation on 2025.10.22 (UT)

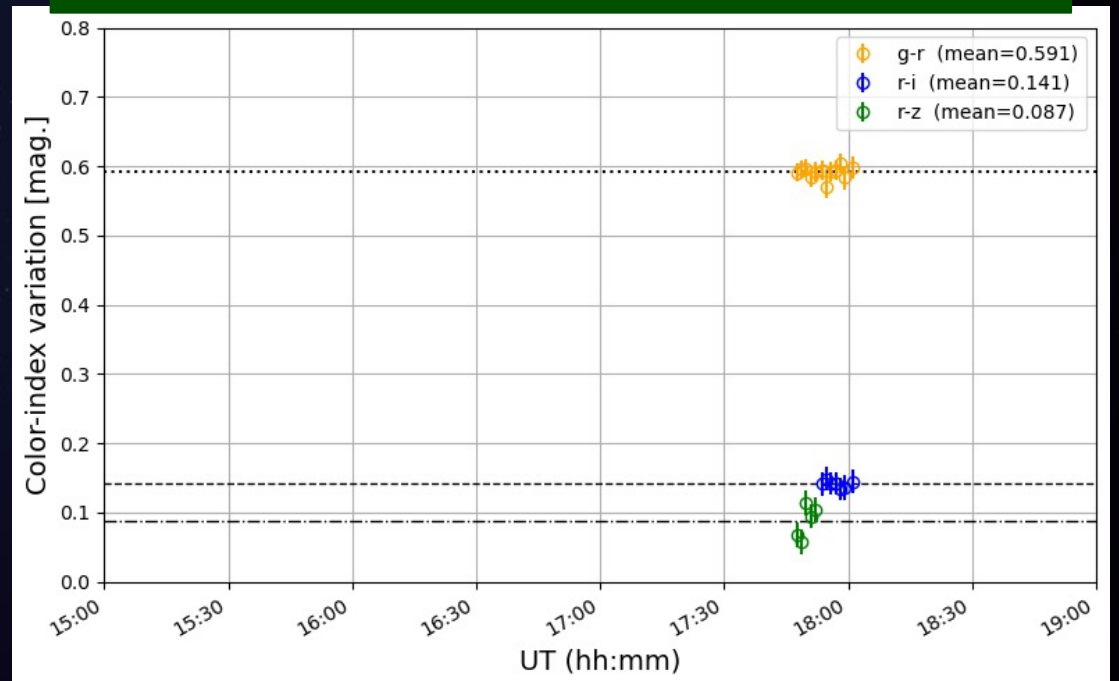


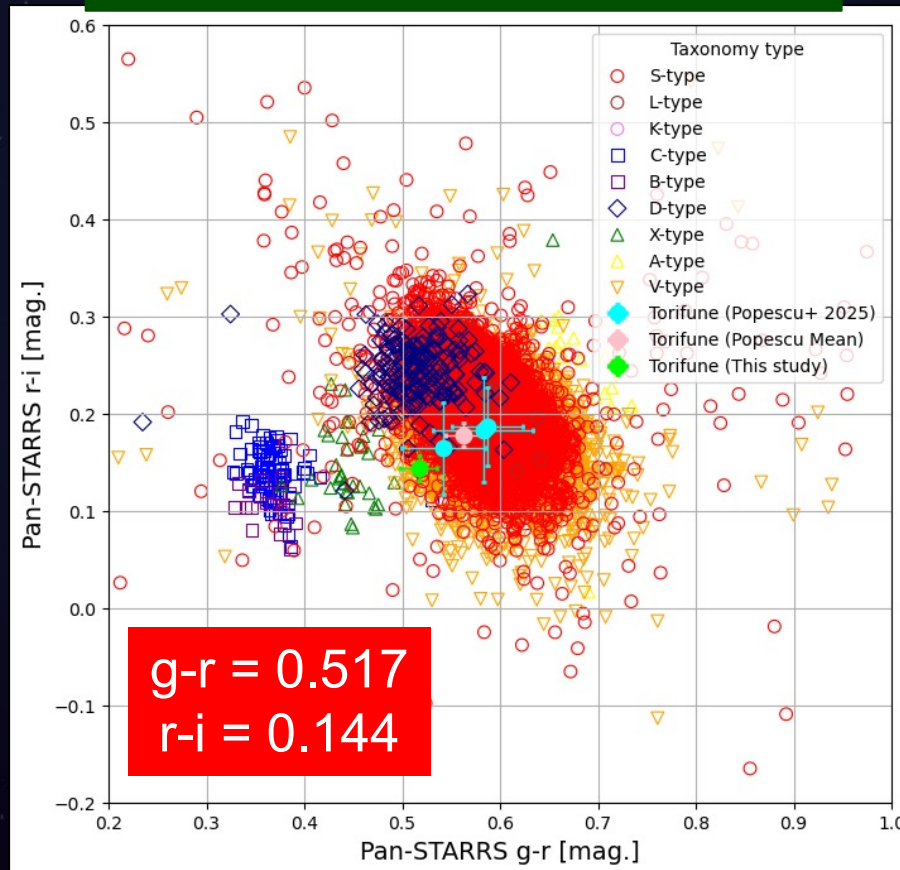
Fig. 71 g-r and r-i color index variation on 11.28 & 10.22 [This study].

5-4. Torifune Observations & Results

【Results】 $r-i$ vs $g-r$ 2-color figure (2024.11.28 & 2025.10.22)

- 2024.11.28 & 2025.10.22
-> Both **S-type**
-> Consistent with Popescu et al., 2025.

Mean $r-i$ vs $g-r$ on 2024.11.28



Mean $r-i$ vs $g-r$ on 2025.10.22

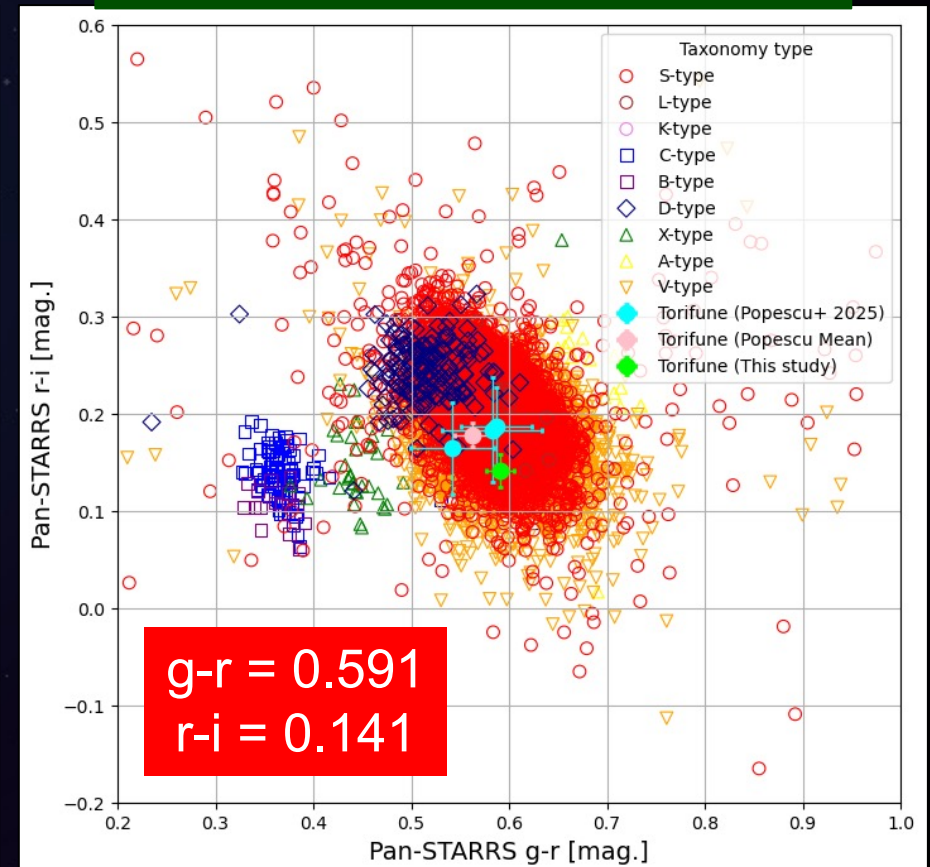


Fig. 72 Pan-STARRS $r-i$ vs $g-r$ 2 color figure on 11.28 & 11.22 [This study].

5-4. Torifune Observations & Results

【Results】 r-z vs g-r 2-color figure (2025.10.22)

- 2025.10.22
 - > **S-type**
 - > Consistent with Popescu et al., 2025.

Mean r-z vs g-r on 2025.10.22

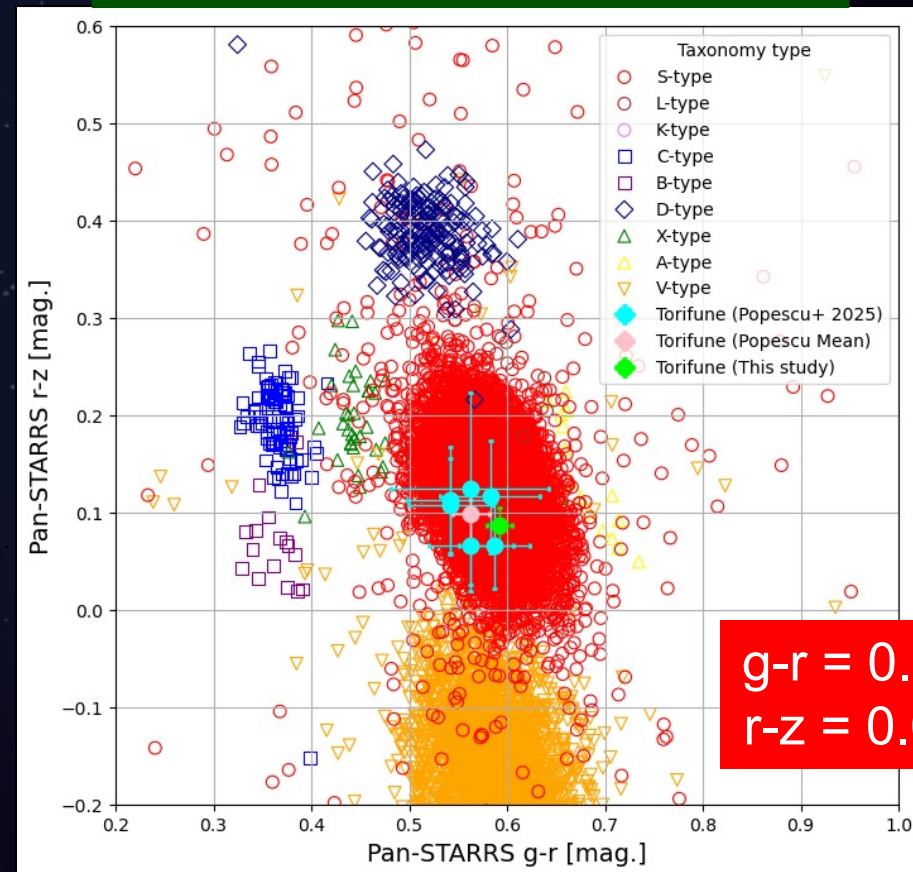


Fig. 73 r-z vs g-r 2 color figure on 10.22 [This study].

5-4. Torifune Observations & Results

【Results】 Phase angle dependence (2024.11.28 & 2025.10.22)

- We could not confirm phase angle dependence of the color index.
-> Consistent with Popescu et al., 2025.
- High accuracy in the color index measurements!

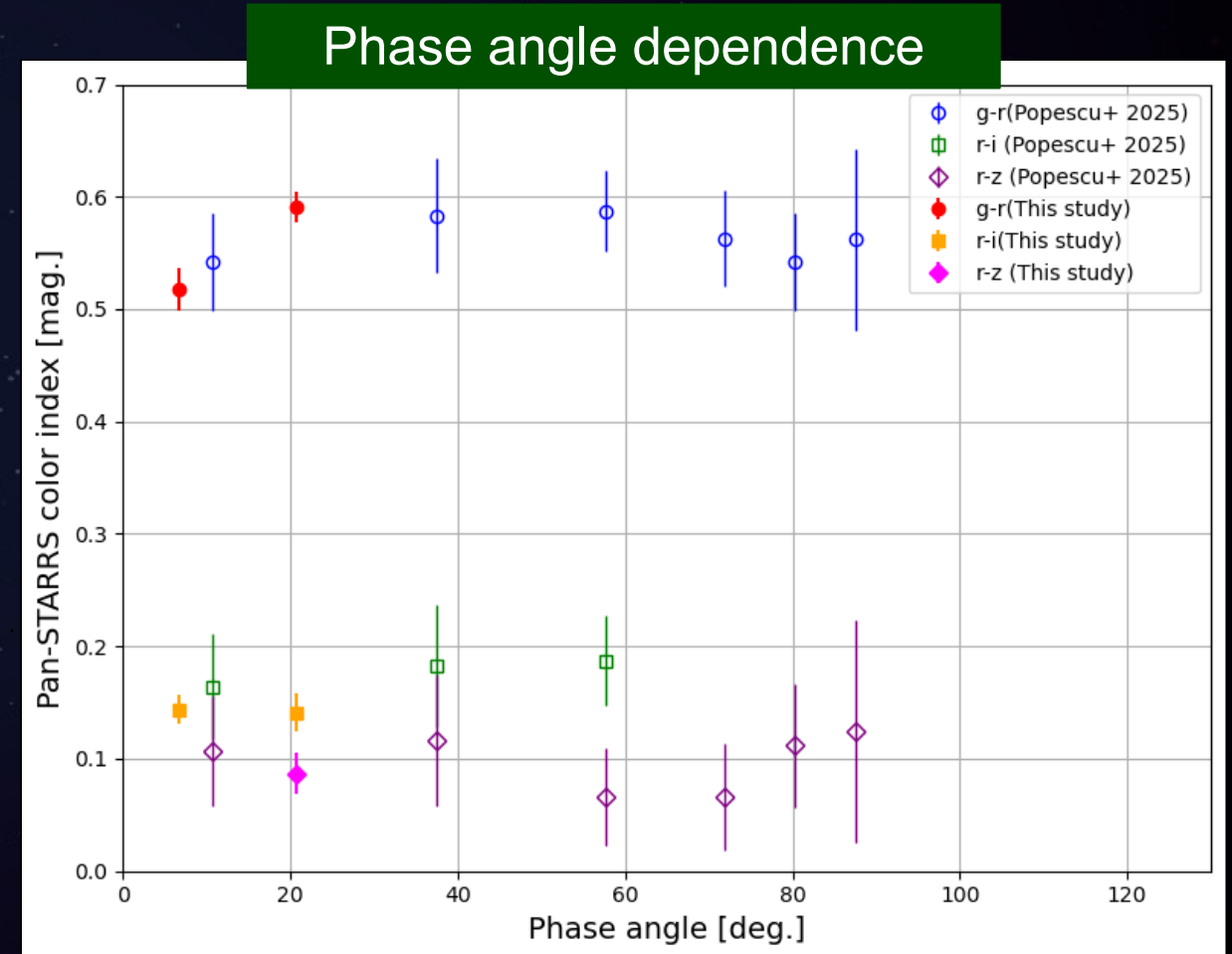


Fig. 74 Phase angle dependence of color index [This study].

5-4. Torifune Observations & Results

【Results】 Consistency with the geometry (2024.11.28)

- Max. cross section -> brightest moment
Min. cross section -> faintest moment
-> **Good consistency with the model!**

Made by Sensyu-san (Co.)!
@Chiba tech univ.

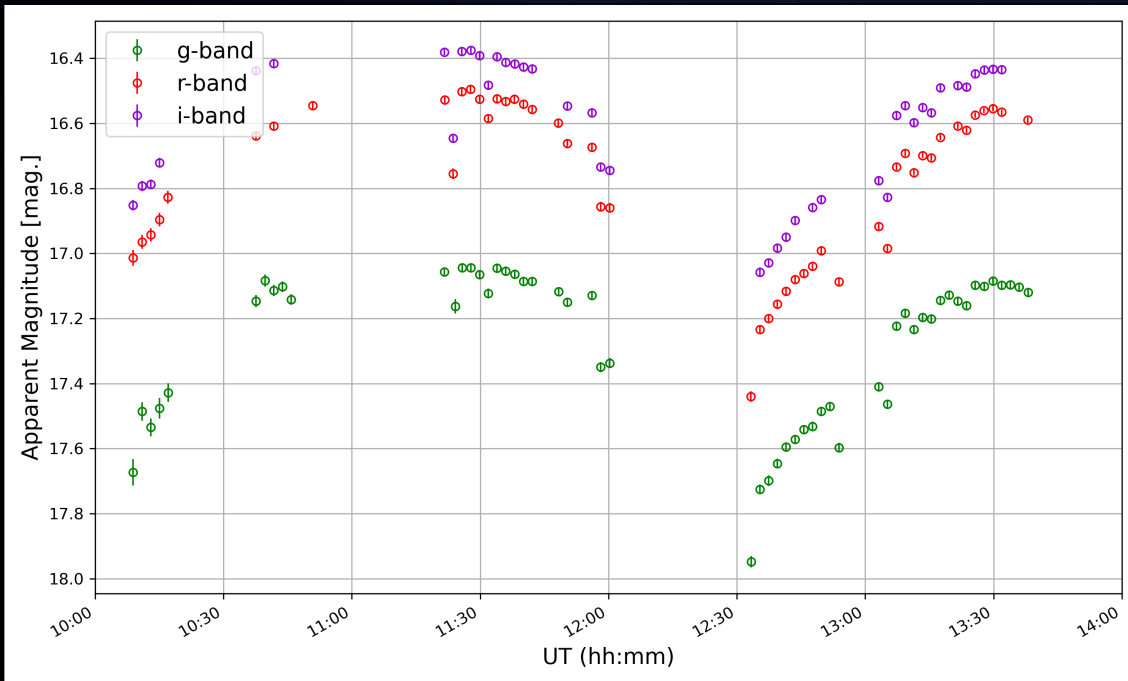
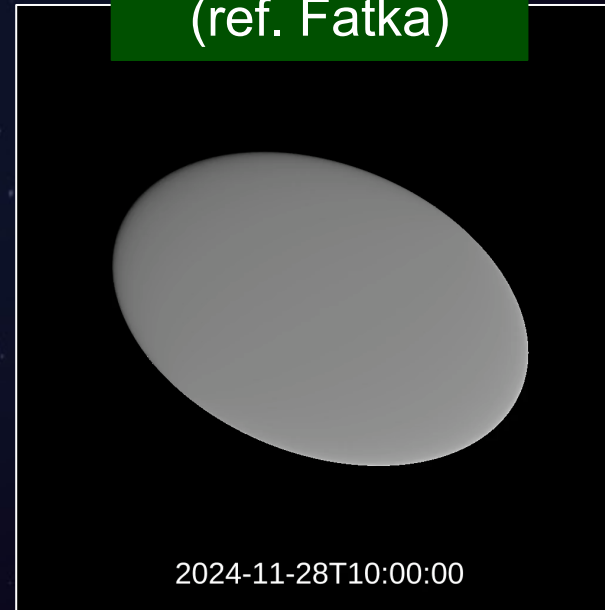


Fig. 75 Multi color light-curve on 11.28 (UT) [This study].

Geometry mov.
(ref. Fatka)



Geometry mov.
(ref. Popescu)

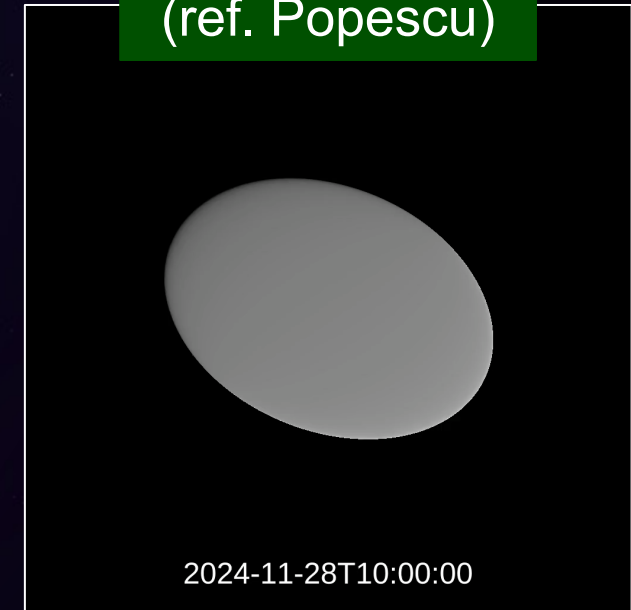


Fig. 76 Torifune geometry on 11.28, reference Fatka et al., 2025 & reference Popescu et al., 2025.

5-4. Torifune Observations & Results

【Results】 Consistency with the geometry (2025.10.22)

- Max. cross section -> brightest moment
Min. cross section -> faintest moment
-> Good consistency with the model!

Made by Sensyu-san (Co.)!
@Chiba tech univ.

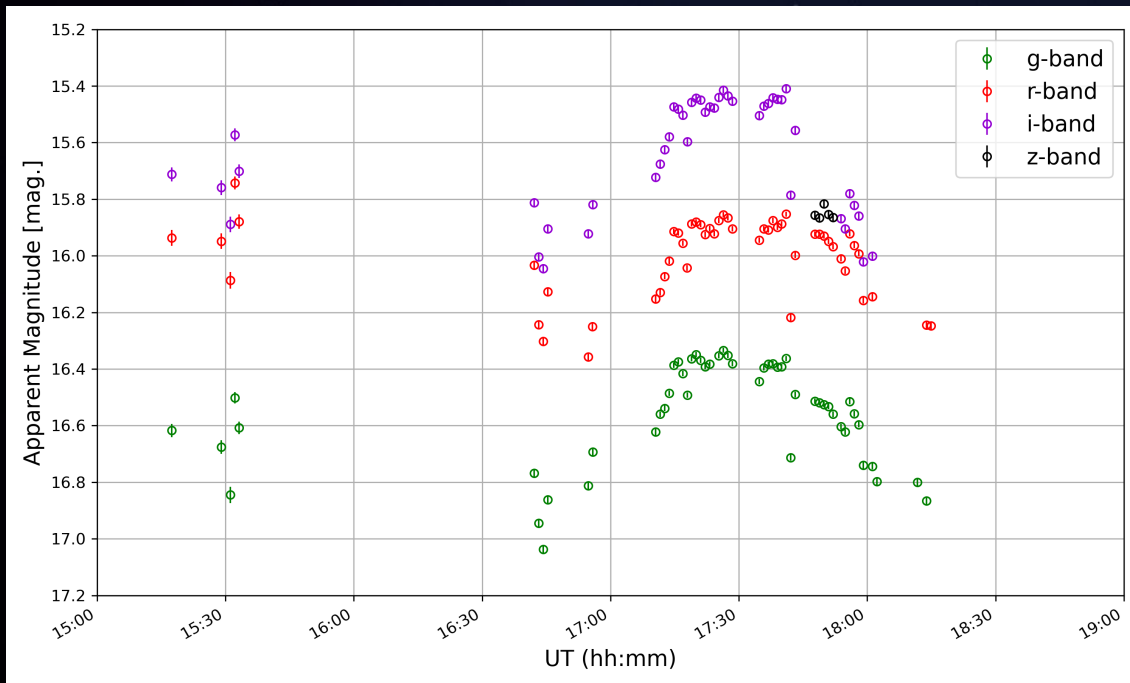
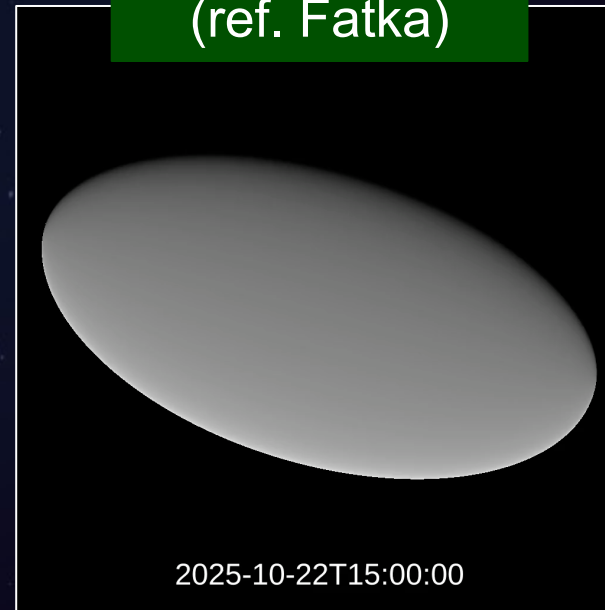


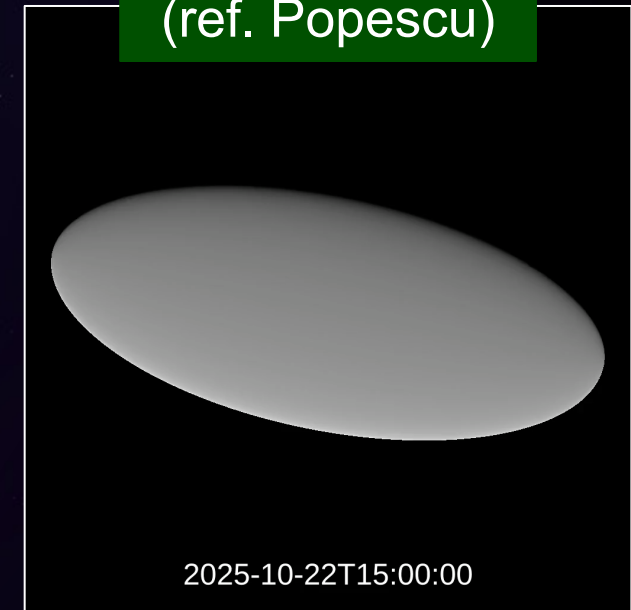
Fig. 77 Multi color light-curve on 11.28 (UT) [This study].

Geometry mov.
(ref. Fatka)



2025-10-22T15:00:00

Geometry mov.
(ref. Popescu)



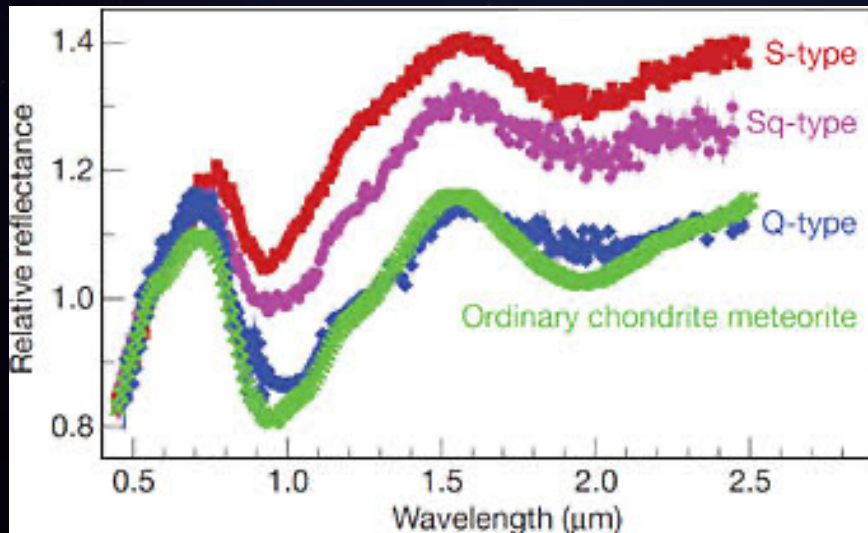
2025-10-22T15:00:00

Fig. 78 Torifune geometry on 10.22, reference Fatka et al., 2025 & reference Popescu et al., 2025.

5-4. Torifune Observations & Results

【Results】 Reflectance spectrum (2025.10.22)

- We transformed color index (g-r, r-i, r-z) to Reflectance; R_g , R_r , R_i , R_z
- Normalized at R_g (central wavelength of the Pan-STARRS g-filter; 0.4849 μm)
- Between S-type and Q-type spectrum
-> **Sq-type!**
-> Consistent with Popescu et al., 2025.



Reflectance spectrum on 2025.10.22

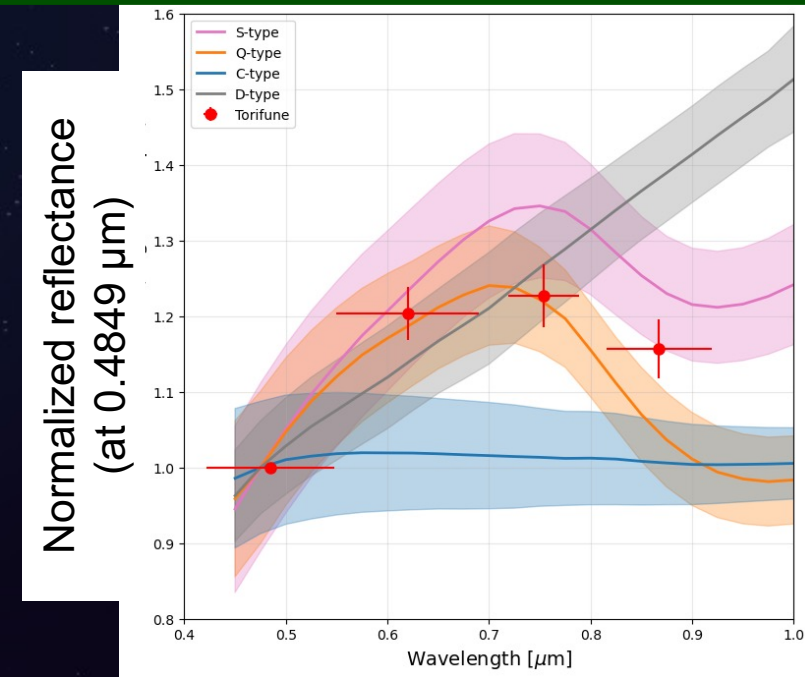


Fig. 79, 80 Reflectance spectrum of each type asteroids, Reflectance spectrum of Torifune on 10.22 [This study].

$$\begin{aligned} R_r &= 10^{-0.4[(r-g)_{3I} - (r-g)_{\odot}]} \\ R_i &= 10^{-0.4[(i-g)_{3I} - (i-g)_{\odot}]} \\ R_z &= 10^{-0.4[(z-g)_{3I} - (z-g)_{\odot}]} \end{aligned}$$

5-5. Torifune Discussion

【Discussion】

- S-type color index, especially Sq-type reflectance spectrum
- Large-scale surface homogeneity
- No phase angle dependence of color



- Consistent with previous studies
- Higher accuracy (photometry; color index) and higher time resolution

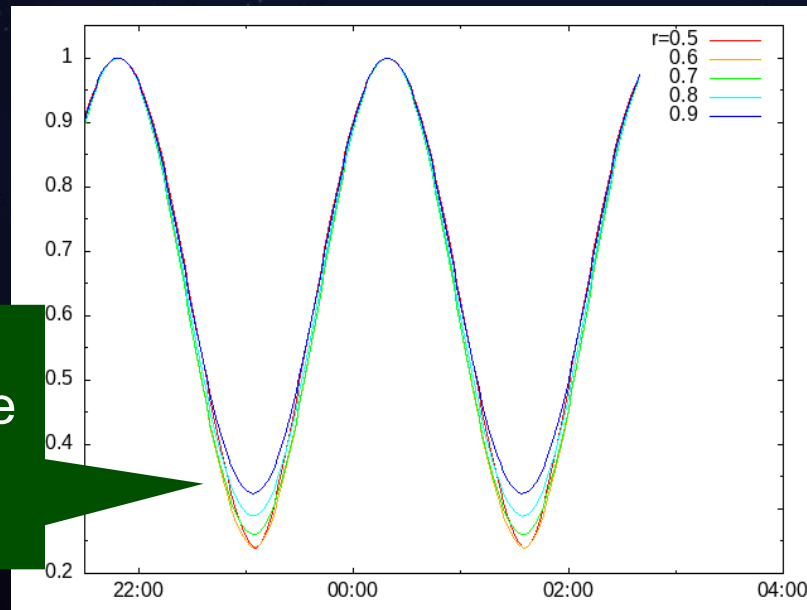
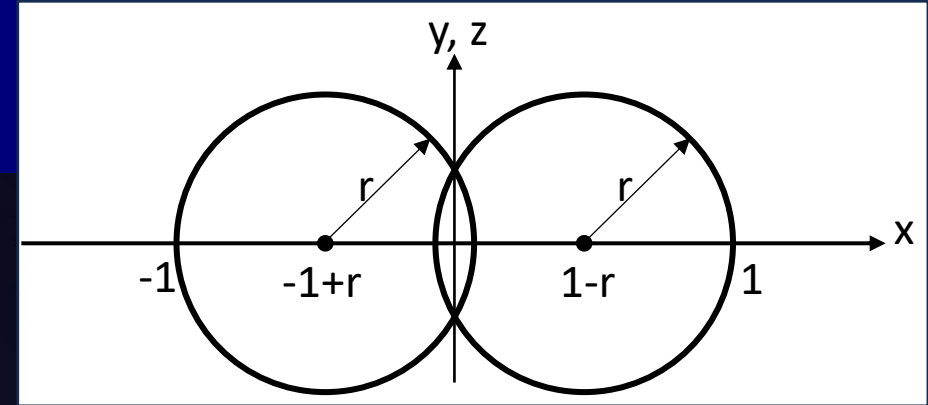


- We can provide valuable results of surface color information for the Hayabusa2# mission.
- We reported results at the Hayabusa2# science meeting (on 7 Oct.).

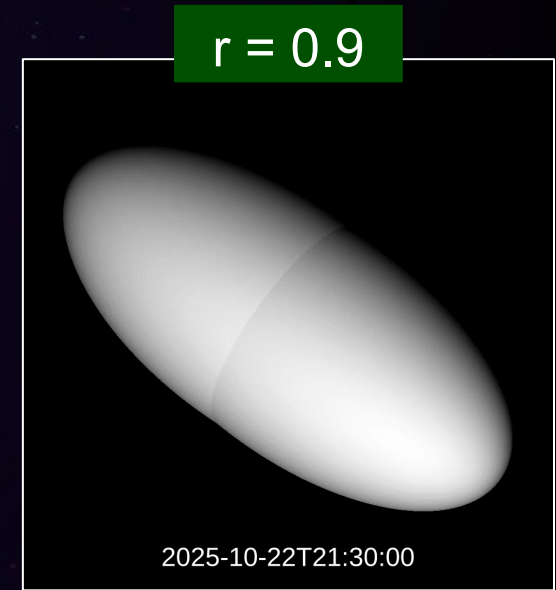
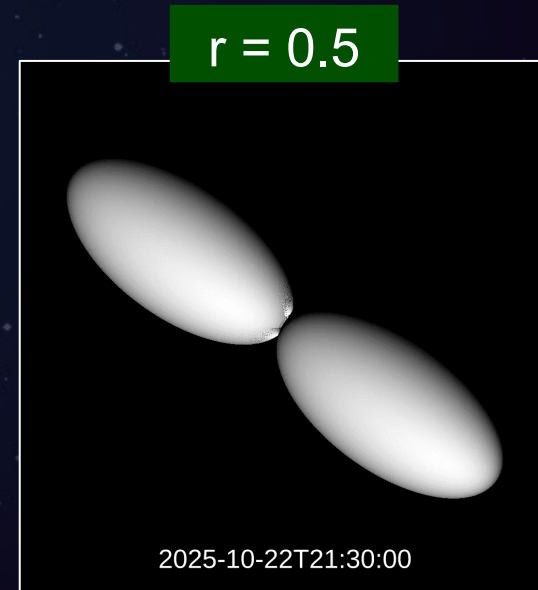
5-6. Torifune Future works

【Future works】

- We will analyze data of 2025.11.06 (but full moon...).
- There is possibility of detecting **contact-binary** or not from light-curve shape.



In the case of a contact-binary, the min. brightness becomes deeper



- We plan to submit a letter paper to the 'Hayabusa2# special issue' in Space Science Reviews (PSJ).

Fig. 81, 82 Simulation light-curve on 10.22, Simulation geometry movie on 10.22.

6. Summary

2024 MK

- S-type
- Surface scattering properties; fine grain size and/or high porosity
- We submitted a letter paper to PASJ on 12 Dec. (waiting now).

Torifune

- S-type, especially Sq-type
- Large-scale surface homogeneity
- No phase angle dependence of color
- Higher accuracy and time resolution than previous studies
- We reported results to the Hayabusa2# team
- We plan to submit a letter paper to the 'Hayabusa2# special issue' in Space Science Reviews (PSJ).

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